LeanForControl

Anand Gokhale Francesco Bullo

University of California, Santa Barbara

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Chapter 1

Analysis toolbox

Facts about real functions, limits and integrals with no differential-equation content of their own. Each section says which later results consume it.

1.1 Continuity toolbox

Two facts about continuous real functions and curves, with no differential-equation content of their own, used throughout the chapters that follow: the comparison lemma needs the last root before a sign change, and the Lyapunov stability arguments need the first time a curve reaches a sphere.

Lemma 1 (Greatest zero of a sign-changing continuous function)**.** *Let g be continuous on $[t_0, t_1]$ with $g(t_0) \le 0 < g(t_1)$. Then g has a* last *root: there is $a \in [t_0, t_1)$ with $g(a) = 0$ and $g(t) > 0$ for every $t \in (a, t_1]$.*

Proof. □

Theorem 2 (First crossing of a sphere)**.** *Let φ be continuous on $[t_0, T]$ with $\|\varphi(t_0) - x_{\mathrm{eq}}\| < \rho$, and suppose some $t_1 \in [t_0, T]$ satisfies $\rho \le \|\varphi(t_1) - x_{\mathrm{eq}}\|$. Then there is a* first *hitting time $\tau \in [t_0, T]$ of the sphere of radius ρ:*

$$\|\varphi(\tau) - x_{\mathrm{eq}}\| = \rho, \qquad \|\varphi(s) - x_{\mathrm{eq}}\| \le \rho \quad \forall\, s \in [t_0, \tau].$$

Proof. Take the minimum of the closed set $[t_0, T] \cap \|\varphi(\cdot) - x_{\mathrm{eq}}\|^{-1}\{\rho\}$, which is nonempty by the intermediate value theorem. Minimality rules out an earlier excursion outside the ball: one would force a second crossing before τ, again by the intermediate value theorem. □

1.2 Limits superior

Real-valued lim sup is an infimum of upper bounds, so it is only well-behaved where the function is bounded. The facts collected here are stated over an arbitrary filter and have no differential-equation content; the Dini derivative of the next section is the only current consumer.

Lemma 3 (Limsup is unchanged by adding a null term)**.** *Let l be a nontrivial filter, let f be bounded above and below along l, and let $g \to 0$ along l. Then $\limsup_l (f + g) = \limsup_l f$.*

Proof. One inequality is subadditivity of lim sup together with $\limsup_l g = 0$. For the other, write $f = (f+g) + (-g)$ and apply the same argument, $-g$ also tending to 0; the boundedness of $f+g$ needed for this follows from that of f and g separately. □

1.3 Dini derivatives

The upper right Dini derivative replaces the ordinary derivative where a function is only known to be continuous, which is the situation the comparison lemma is in. It is defined here as a lim sup valued in $\mathbb{R}$ rather than in $[-\infty, +\infty]$, so most results below carry a hypothesis that the forward difference quotients are bounded: without it an unbounded quotient makes the lim sup an infimum over the empty set, which $\mathbb{R}$ reports as 0.

Definition 4 (Upper right Dini derivative)**.** The *upper right Dini derivative* of $f : \mathbb{R} \to \mathbb{R}$ at t is

$$D^+ f(t) \;=\; \limsup_{h \to 0^+} \frac{f(t+h) - f(t)}{h}.$$

When f is differentiable at t this equals $f'(t)$.

1.3.1 Relationship to the classical derivative

Lemma 5 (Forward difference quotients of a differentiable function)**.** *If f has right derivative L at t, then $\frac{f(t+h)-f(t)}{h} \to L$ as $h \to 0^+$.*

Proof. □

Lemma 6 (Dini derivative of a differentiable function)**.** *If f has right derivative L at t, then $D^+ f(t) = L$: the Dini derivative agrees with the classical one wherever the latter exists.*

Proof. □

1.3.2 Characterization by difference quotients

Lemma 7 (Upper bound on the Dini derivative)**.** *Let the forward difference quotients of f at t be bounded above and cobounded below along $h \to 0^+$. Then $D^+ f(t) \le L$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ is eventually at most $L + \varepsilon$ as $h \to 0^+$.*

Proof. □

Lemma 8 (Lower bound on the Dini derivative)**.** *Under the same boundedness conditions as lemma 7, $L \le D^+ f(t)$ if and only if for every $\varepsilon > 0$ the quotient $\frac{f(t+h)-f(t)}{h}$ exceeds $L - \varepsilon$ frequently as $h \to 0^+$.*

Proof. □

1.3.3 Linearity properties

Lemma 9 (Subadditivity of the Dini derivative)**.** *The upper right Dini derivative is subadditive: $D^+(f+g)(t) \le D^+f(t) + D^+g(t)$, provided the forward difference quotients of f are bounded above and below and those of g are bounded above and cobounded below.*

Proof. □

Lemma 10 (Adding a differentiable function)**.** *If the forward difference quotients of f at t are bounded above and below, and g is differentiable at t, then $D^+(f+g)(t) = D^+f(t) + g'(t)$. Subadditivity becomes equality once one summand has a genuine derivative.*

Proof. □

Lemma 11 (Adding a linear function)**.** *If the forward difference quotients of f at t are bounded above and below, then $D^+(f + c\,\mathrm{id})(t) = D^+f(t) + c$ — the case $g(s) = cs$ of lemma 10, and the form the comparison arguments use.*

Proof. □

Lemma 12 (Positive scalar multiple)**.** *For $c > 0$, and forward difference quotients of f at t bounded above and cobounded below, $D^+(cf)(t) = c\,D^+f(t)$. Positivity is essential: a negative constant reverses the order and turns the* lim sup *into a* lim inf*.*

Proof. Multiplication by $c > 0$ is an order isomorphism of $\mathbb{R}$, and a lim sup commutes with one. Going through the order isomorphism rather than through the defining infimum avoids the degenerate empty-inf case. □

Lemma 13 (Negation and the lower Dini derivative)**.** *If the forward difference quotients of f at t are bounded above and below, then $-D^+(-f)(t) \le D^+f(t)$. The left-hand side is the* lower *right Dini derivative, so this is the expected inequality $D_+f(t) \le D^+f(t)$ written in terms of D^+ alone.*

Proof. □

1.3.4 The comparison step

Lemma 14 (Dini derivative from a one-sided bound)**.** *Let $a < b$, let $z(a) = v(a)$ and $z(t) < v(t)$ for all $t \in (a, b]$, let z have right derivative d_z at a, and let the forward difference quotients of v at a be bounded above. Then $d_z \le D^+v(a)$.*

Proof. On $(0, b-a)$ the hypotheses give $z(a+h) - z(a) \le v(a+h) - v(a)$, so the forward quotient of z is eventually below that of v. The quotient of z converges to d_z, so d_z = lim sup of it, and lim sup is monotone. □

1.4 Interval integrals

Estimates and integrability side conditions for the interval integral, used wherever a differential equation is handled in integral form. The last three concern an integral whose endpoints move: normalizing by the window length makes it a mean value, after which continuity, antitonicity and decay each need only a hypothesis on the endpoints themselves. They support the class $\mathcal{KL}$ construction of the chapter on comparison functions.

Lemma 15 (Difference of integrals over adjacent intervals)**.** *For μ continuous on $[a,t]$ and $s \in [a,t]$,*

$$\int_a^t \mu - \int_a^s \mu = \int_s^t \mu.$$

Proof. □

Lemma 16 (Integral bound from a pointwise multiplicative bound)**.** *Let u take values in a normed space and v be real, both integrable on $[a,b]$ with $a \le b$. If $\|u(s)\| \le L\, v(s)$ for all $s \in [a,b]$, then*

$$\Big\| \int_a^b u \Big\| \le L \int_a^b v.$$

Proof. □

Lemma 17 (Integral bound from a constant pointwise bound)**.** *If $\|u(s)\| \le C$ for all $s \in [a,b]$ with $a \le b$, then*

$$\Big\| \int_a^b u \Big\| \le C\,(b-a).$$

Proof. □

Lemma 18 (Integral of a constant)**.** $\int_a^b C \,\mathrm{d}s = (b-a)\,C$ *for a real constant C.*

Proof. □

Lemma 19 (Interval integrability of a continuous composition)**.** *If f is jointly continuous and z is continuous on the segment between t_0 and t_1, then $s \mapsto f(s, z(s))$ is interval-integrable between them. This is the integrability side condition every integral-form solution argument needs.*

Proof. □

Lemma 20 (Differentiating an integral in its upper limit)**.** *For μ continuous on $[a,b]$ and $t \in (a,b)$,*

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big|_{x=t} \int_a^x \mu(\tau)\,\mathrm{d}\tau \;=\; \mu(t).$$

Proof. □

Lemma 21 (Continuity of an integral in its upper limit)**.** *If f is integrable on $[a,t]$ with $a \le t$, then $s \mapsto \int_a^s f(\tau)\,\mathrm{d}\tau$ is continuous on $[a,t]$.*

Proof. □

Lemma 22 (Continuity of an integral in both endpoints)**.** *Let f be locally integrable on $(0,\infty)$, and let a, b be continuous on a set S with $a(\eta), b(\eta) > 0$ in S. Then*

$$\eta \;\longmapsto\; \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is continuous on S.

Proof. □

Lemma 23 (The sliding average of an antitone function is antitone)**.** *Let f be antitone and locally integrable on $(0,\infty)$, and let a, b be monotone on a set S with $0 < a(\eta) < b(\eta)$. Then the mean value*

$$\eta \;\longmapsto\; \frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x$$

is antitone on S.

Proof. □

Lemma 24 (Sliding average of an eventually vanishing function)**.** *Let f be nonneg, antitone and locally integrable on $(0,\infty)$ with $f(s) \to 0$ as $s \to +\infty$, and let $a(\eta) < b(\eta)$ with $a(\eta) \to +\infty$ as $\eta \to +\infty$. Then*

$$\frac{1}{b(\eta)-a(\eta)} \int_{a(\eta)}^{b(\eta)} f(x)\,\mathrm{d}x \;\longrightarrow\; 0 \qquad (\eta \to +\infty).$$

The mean is squeezed between 0 and $f(a(\eta))$; no hypothesis relating a to b is needed.

Proof. □

1.5 Positive definite functions

Positive definiteness in the sense of functions, not matrices: a function that vanishes at a point and is positive elsewhere. It is the shape every Lyapunov function and every comparison bound takes.

Definition 25 (Positive definite function)**.** A function W is *positive definite* on S about x_0 when it is continuous on S, $W(x_0) = 0$, and $W(x) > 0$ for every $x \in S \;\; \{x_0\}$.

Chapter 2

Ordinary differential equations

2.1 Gronwall–Bellman inequality

A function obeying an integral inequality is bounded by the solution of the corresponding integral equation. The two specializations are what the rest of the development actually calls: constant Λ gives exponential growth, and constant Λ and μ gives the $Ce^{\mu(t-a)}$ bound behind every continuous-dependence estimate.

Theorem 26 (Gronwall–Bellman inequality)**.** *Khalil, Lemma A.1. Let $\Lambda, \mu : [a,b] \to \mathbb{R}$ be continuous with $\mu \geq 0$, and let $y : [a,b] \to \mathbb{R}$ be continuous satisfying*

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then

$$y(t) \;\leq\; \Lambda(t) + \int_a^t \Lambda(s)\, \mu(s)\, e^{\int_s^t \mu(\tau)\,\mathrm{d}\tau}\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Proof. An integrating-factor argument. Write $z(t) = \int_a^t \mu y$, $M(t) = \int_a^t \mu$, and $v = z + \Lambda - y$, so that $v \geq 0$ is exactly the hypothesis. The transform $w = e^{-M} z$ has, by the product rule and lemma 20,

$$\dot{w} = e^{-M}\mu\,(y - z) = e^{-M}\mu\,(\Lambda - v) \;\leq\; e^{-M}\mu\Lambda,$$

the inequality because e^{-M}, μ and v are all nonnegative — this is the only place $\mu \geq 0$ is used, and it is essential: a sign change would reverse it. Integrating from a, where $w(a) = 0$, gives $w(t) \leq \int_a^t e^{-M(s)}\mu(s)\Lambda(s)\, ds$. Multiplying by $e^{M(t)}$ and folding $e^{M(t)}e^{-M(s)} = e^{\int_s^t \mu}$ turns this into the stated bound on $z(t)$, and $y \leq \Lambda + z$ finishes. □

Lemma 27 (Gronwall–Bellman with a constant rate)**.** *Khalil, Lemma A.1 (the case $\Lambda(t) \equiv \Lambda$ constant). Let μ be continuous and nonnegative on $[a,b]$, and let y be continuous with*

$$y(t) \;\leq\; C + \int_a^t \mu(s)\, y(s)\, \mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then $y(t) \leq C\, e^{\int_a^t \mu(\tau)\,\mathrm{d}\tau}$ on $[a,b]$.

Proof. theorem 26 at $\Lambda \equiv C$ gives $y(t) \le C + \int_a^t C\mu(s)e^{\int_s^t \mu}\,ds$, and the integral evaluates in closed form: $s \mapsto -C\,e^{M(t)-M(s)}$ is an antiderivative of the integrand, so the integral is $Ce^{M(t)} - C$. The two C's cancel. □

Lemma 28 (Gronwall–Bellman with constant data)**.** *Khalil, Lemma A.1 (the case Λ and $\mu \ge 0$ both constant). Let $\mu \ge 0$ be constant and let y be continuous with*

$$y(t) \;\le\; C + \int_a^t \mu\, y(s)\,\mathrm{d}s \qquad \forall\, t \in [a,b].$$

Then $y(t) \le C\,e^{\mu(t-a)}$ on $[a,b]$. This is the form the continuous-dependence estimates use, and the reason their bounds carry a factor $e^{L(t-t_0)}$.

Proof. lemma 27 with $\int_a^t \mu = \mu(t-a)$. □

2.2 Existence, uniqueness and continuous dependence

Definition 29 (Integral solution)**.** A function x, defined on the segment between t_0 and t_1, is an *integral solution* of $\dot{x} = F(t,x)$ with initial value x_0 when

$$x(t) \;=\; x_0 + \int_{t_0}^t F(s,\,x(s))\,\mathrm{d}s$$

for every t between t_0 and t_1.

Lemma 30 (Re-anchoring an integral solution)**.** *Let x be a continuous integral solution of $\dot{x} = F(t,x)$ with initial value x_0 on the segment between t_0 and t_1, with F jointly continuous. Then for any s, r on that segment, x is an integral solution on the segment between s and r, with initial value $x(s)$:*

$$x(t) \;=\; x(s) + \int_s^t F(w, x(w))\,\mathrm{d}w \qquad \textit{for } t \textit{ between } s \textit{ and } r.$$

Leaving both endpoints free means the domain need not shrink — taking $s = t_1$ and $r = t_0$ re-anchors at one end of the original segment while retaining all of it.

Proof. Subtract the defining equations at t and at s and split the integral at s, which is legitimate because $w \mapsto F(w, x(w))$ is interval-integrable on each piece. □

Lemma 31 (Time reflection of an integral solution)**.** *Let x be an integral solution of $\dot{x} = F(t,x)$ with initial value x_0 on the segment between t_0 and t_1. Then $\sigma \mapsto x(t_0 + t_1 - \sigma)$ is an integral solution, with the same initial value x_0, of the reflected field*

$$(r, y) \;\longmapsto\; -F(t_0 + t_1 - r,\; y)$$

on the segment between t_1 and t_0.

The map $\sigma \mapsto t_0 + t_1 - \sigma$ is an involution exchanging the two endpoints, so reflection turns a statement anchored at t_0 into one anchored at t_1 and nothing more. Its purpose is to let an argument that is inherently forward-marching — a Grönwall bootstrap, say — be applied unchanged in backward time.

Proof. Substituting $r \mapsto t_0 + t_1 - r$ in the defining integral reverses the orientation of the interval, contributing one sign, and the negation of the field contributes another; the two cancel. □

2.2.1 The integral and differential formulations

An ODE can be written as a differential equation, $\dot{x} = F(t, x)$, or as the equivalent integral equation. The continuous-dependence and Gronwall arguments want the integral form, which presupposes no differentiability and carries its initial condition; the Lyapunov arguments want the differential form, so that $V \circ \varphi$ can be differentiated. The results below are the two directions of the fundamental theorem of calculus relating them.

They are provision rather than current practice: nothing in the stability chapters has yet been routed through this equivalence, so it is the intended path from a trajectory to the results of this chapter rather than an exercised one.

Lemma 32 (Integral curves are integral solutions)**.** *Let x be an integral curve of F on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral solution on that segment, anchored at its own value $x(t_0)$:*

$$x(t) \;=\; x(t_0) + \int_{t_0}^{t} F(s, x(s))\,\mathrm{d}s.$$

Continuity of x itself is not assumed — it follows from differentiability.

Proof. The fundamental theorem of calculus. The derivative hypothesis holds on an open neighbourhood of each interior point, so it yields the right-derivative form the theorem needs, and continuity of F along x gives the integrability side condition. □

Lemma 33 (Integral solutions are integral curves)**.** *Let x be an integral solution of $\dot{x} = F(t, x)$ on the segment between t_0 and t_1, with $s \mapsto F(s, x(s))$ continuous there. Then x is an integral curve of F on that segment: it is differentiable, with $\dot{x}(t) = F(t, x(t))$.*

Proof. Differentiating the primitive: the integral term has derivative $F(t, x(t))$ by the fundamental theorem of calculus, and the constant x_0 contributes nothing. Transport that along the defining equation, which says x agrees with the primitive. □

Lemma 34 (Integral and differential forms agree)**.** *Given continuity of $s \mapsto F(s, x(s))$ on the segment between t_0 and t_1, the two formulations agree: x is an integral solution anchored at $x(t_0)$ if and only if it is an integral curve of F there. Continuity of F along x is exactly the price of the equivalence — without it the integral form is the weaker notion.*

Proof. The two implications are lemma 33 and lemma 32. □

Lemma 35 (Integral and differential forms agree, forward time)**.** *lemma 34 in forward time: for $t_0 \le t_1$, the equivalence holds over $[t_0, t_1]$. This is the interval the finite-forward stability predicates are phrased over, so this is the form the stability track would use to reach the results of this chapter.*

Proof. □

Lemma 36 (Integral form for an autonomous field)**.** *Let g be a continuous* autonomous *field and let x be an integral curve of g on $[t_0, t_1]$, $t_0 \le t_1$. Then x is an integral solution there, anchored at $x(t_0)$.*

This is the autonomous, forward-time specialization of lemma 35, and so the shape a trajectory of an autonomous system has: continuity of the field is the only hypothesis, the continuity of x along which F must be continuous being supplied by the curve itself.

Proof. □

2.2.2 Time invariance and local existence

A time-invariant field has no preferred time origin: a solution can be shifted to start at time 0.

Lemma 37 (Time invariance)**.** *If x is an integral curve of an autonomous field g on a set s, then $t \mapsto x(t + \Delta t)$ is an integral curve of the same g on $s - \Delta t$, for every Δt.*

An autonomous field has no preferred time origin, so translating a solution translates its interval of definition and changes nothing else. For a time-varying field the translated solution would solve the translated *equation instead.*

Proof. □

Lemma 38 (Shifting a solution segment to the origin)**.** *lemma 37 in the form the finite-segment predicates need: a solution segment of an autonomous field on $[t_0, t_1]$ becomes one on $[0, t_1 - t_0]$ under $s \mapsto x(s + t_0)$.*

This is why the autonomous stability predicates may fix the initial time at 0 without loss: any segment can be shifted there.

Proof. □

Lemma 39 (Local existence for a C^1 field)**.** *Let g be an autonomous field that is C^1 at x_0. Then there is a $T > 0$ and a curve φ with $\varphi(0) = x_0$ that is an integral curve of g on $[0, T]$: a nontrivial solution segment exists from every point at which the field is C^1.*

The anchor 0 is a construction choice rather than a restriction — by lemma 37 a segment from any other anchor follows.

Reference: the Picard–Lindelöf local existence theorem.

Proof. Picard–Lindelöf local existence: being C^1 at x_0 gives a closed ball on which g is Lipschitz, and hence a solution on some $(-\varepsilon, \varepsilon)$; take $T := \varepsilon/2$. □

Theorem 40 (Picard–Lindelöf on a compact interval)**.** *Let $f : \mathbb{R} \to E \to E$ be jointly continuous and K-Lipschitz in the state variable, uniformly in time, on a Banach space E. Then for every $t_0 \le t_1$ and every $x_0 \in E$ there is an α with $\alpha(t_0) = x_0$ solving $\dot{\alpha} = f(t, \alpha)$ on all of $[t_0, t_1]$.*

Global — rather than merely local — Lipschitz continuity is what makes the solution reach t_1: it forces linear growth in the state, ruling out finite-time blowup.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2 (Global Existence and Uniqueness), proved in his Appendix C.1. Difference: Khalil assumes f piecewise continuous in t; here it is jointly continuous.*

Proof. Mathlib's Picard–Lindelöf theorem supplies a solution on a time interval of length Δt only when $\|f\| \le L$ on a ball of radius a about x_0 with $L\,\Delta t \le a$. For a K-Lipschitz field the sharpest available bound on that ball is $L = M + Ka$, where M bounds $\|f(t, x_0)\|$, so the condition becomes $M\Delta t \le a(1 - K\Delta t)$ — solvable in a exactly when $K\Delta t < 1$. One application therefore never covers a long interval.

Take instead a step h with $Kh \le 1/2$; then $a = 2Mh$ and $L = 2M$ satisfy the hypotheses, since $\|f(t, x)\| \le M + Ka = M(1 + 2Kh) \le 2M$ and $Lh = 2Mh = a$. This gives a solution on $[s, s + h]$ from any initial state. Chain n such steps, splicing consecutive segments at their shared endpoint: away from a junction the glued curve agrees with one piece on a neighbourhood, and at a junction the two one-sided derivatives combine. Choosing n with $nh \ge t_1 - t_0$ covers $[t_0, t_1]$. □

Lemma 41 (Right derivatives of an integral curve)**.** *If α is an integral curve of f on $[t_0, t_1]$, then for every $s \in [t_0, t_1)$ it has right derivative $f(s, \alpha(s))$ at s, i.e. α has derivative $f(s, \alpha(s))$ within $[s, \infty)$ at s.*

Proof. □

Lemma 42 (Gronwall separation of two solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$. Then*

$$d(\alpha(t), \beta(t)) \le d(\alpha(t_0), \beta(t_0))\, e^{K(t-t_0)} \qquad \forall\, t \in [t_0, t_1].$$

Uniqueness (lemma 43) is the case where the initial distance is zero; continuous dependence on the initial state is the statement read as a bound on the whole interval.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.4, in the unperturbed case: two solutions of the* same *field, so the perturbation bound is zero.*

Proof. □

Lemma 43 (Uniqueness of solutions)**.** *Let f be K-Lipschitz in the state variable, uniformly in time, and let α, β be integral curves of f on $[t_0, t_1]$ with $\alpha(t_0) = \beta(t_0)$. Then $\alpha = \beta$ on $[t_0, t_1]$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), the uniqueness half of Theorem 3.2.*

Proof. The initial distance in lemma 42 is zero, so the bound forces the distance to vanish throughout. □

Theorem 44 (Picard–Lindelöf on the forward ray)**.** *Let f be jointly continuous and K-Lipschitz in the state variable, uniformly in time. Then for every t_0 and x_0 there is an α with $\alpha(t_0) = x_0$ solving $\dot\alpha = f(t, \alpha)$ on all of $[t_0, \infty)$, and any other such solution agrees with it on $[t_0, \infty)$.*

Uniqueness is agreement on $[t_0, \infty)$, not equality of functions: the defining condition says nothing about $t < t_0$, so solutions may differ there.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, whose t_1 is arbitrary, extended to the forward ray.*

Proof. For each n take a solution α_n on $[t_0, t_0 + n]$ (theorem 40). By uniqueness (lemma 43) these agree wherever two of them are both defined, so $\Phi(t) := \alpha_{\lceil t - t_0 \rceil}(t)$ is well defined and agrees with α_n on all of $[t_0, t_0 + n]$. Near any $t \ge t_0$ choose $n > t - t_0$; then $[t_0, t_0 + n]$ is a neighbourhood of t within $[t_0, \infty)$ on which $\Phi = \alpha_n$, so Φ inherits the derivative there. Uniqueness on the ray follows by applying the compact-interval uniqueness at each n. □

Theorem 45 (Picard–Lindelöf, scalar integral form)**.** *Let $g : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ be jointly continuous and L-Lipschitz in its state argument, uniformly in time, and let $t_0 \le t_1$. Then for every x_0 there is a z, continuous on $[t_0, t_1]$, that is an integral solution of $\dot z = g(t, z)$ with $z(t_0) = x_0$ and has right derivative $g(s, z(s))$ at each $s \in [t_0, t_1)$.*

This is the scalar specialization of theorem 40; it discharges the perturbed-solution hypothesis that theorem 49 takes for generality.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 3.2, scalar case.*

Proof. Apply theorem 40 with $E = \mathbb{R}$. Continuity on $[t_0, t_1]$ is automatic for an integral curve, which converts the differential form into the integral one (lemma 35); the right derivatives are lemma 41. □

2.2.3 Continuous dependence

Two solutions of the same equation, started at different points or subject to different perturbations, separate no faster than exponentially — at a rate fixed by the Lipschitz constant of the field. Everything downstream that needs uniqueness gets it from here, by taking the two solutions to agree initially and the perturbation to vanish, which collapses the bound to zero.

Theorem 46 (Continuous dependence on initial states and parameters)**.** *Khalil, Theorem 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on the segment between t_0 and t_1, let y be an integral solution of $\dot{y} = f(t, y)$ with $y(t_0) = y_0$, and let z be an integral solution of the perturbed equation $\dot{z} = f(t, z) + g(t, z)$ with $z(t_0) = z_0$, where $\|g(t, x)\| \le \mu$. Then for every t on that segment,*

$$\|y(t) - z(t)\| \;\le\; \|y_0 - z_0\|\, e^{L|t-t_0|} \;+\; \frac{\mu}{L}(e^{L|t-t_0|} - 1).$$

Two solutions may separate, but no faster than exponentially, at a rate set by the Lipschitz constant alone; the two error sources — a different starting point and a perturbed field — contribute additively. The bound is stated on the unordered segment, so it runs in either time direction.

The perturbation g is assumed only bounded and integrable along z, never continuous, so z need not be differentiable anywhere. That is why the theorem is phrased over the integral formulation rather than the differential one.

Proof. The forward case is a Grönwall bootstrap. Subtracting the two integral equations and applying the triangle inequality gives

$$\|y(\tau) - z(\tau)\| \;\le\; \|y_0 - z_0\| + \mu(\tau - t_0) + L\int_{t_0}^{\tau} \|y(s) - z(s)\|\,\mathrm{d}s,$$

the Lipschitz hypothesis bounding the f-difference and the uniform bound on g the rest. Adding μ/L to both sides absorbs the linear term $\mu(\tau - t_0)$ into the integral, putting the inequality in the shape lemma 28 expects, and Grönwall delivers the exponential.

That bootstrap is inherently forward-marching, so rather than repeat it for the backward case, reflect: lemma 31 sends $\sigma \mapsto t_0 + t_1 - \sigma$ and negates the field, turning the backward instance into a forward one on $[t_1, t_0]$ to which the same argument applies unchanged. □

Theorem 47 (Continuous dependence on parameters)**.** *Khalil, Theorem 3.5. If y solves $\dot{y} = f(t, y)$ and z solves $\dot{z} = f(t, z) + g(t, z)$ with $\|g(t, x)\| \le \alpha$ and $\|z_0 - y_0\| \le \alpha$, and $\alpha(1 + 1/L)e^{L(t_1-t_0)} \le \varepsilon$, then $\|y(t) - z(t)\| \le \varepsilon$ for all $t \in [t_0, t_1]$.*

Proof. Specialise theorem 46 with $\mu = \alpha$ and $\|y_0 - z_0\| \le \alpha$: for $t \in [t_0, t_1]$ it gives $\|y(t) - z(t)\| \le \alpha e^{L(t-t_0)} + \frac{\alpha}{L}(e^{L(t-t_0)} - 1) = \alpha(1 + 1/L)e^{L(t-t_0)} - \alpha/L$. Drop the $-\alpha/L$ and bound $e^{L(t-t_0)}$ by $e^{L(t_1-t_0)}$ to get at most ε. □

2.3 Comparison lemma

The comparison lemma bounds a function satisfying a differential *inequality* by the solution of the corresponding differential *equation*. The inequality is stated with the Dini derivative, so v need only be continuous — which is what makes the lemma applicable to $\|x(t)\|$, not differentiable where a trajectory crosses the origin.

Lemma 48 (Integral form from a pointwise derivative)**.** *Let u be continuous on $[t_0, t_1]$ with $u(t_0) = u_0$ and $\dot{u}(t) = f(t, u(t))$ at every interior point, and let f be jointly continuous. Then u is an integral solution:*

$$u(s) = u_0 + \int_{t_0}^{s} f(\tau, u(\tau))\, \mathrm{d}\tau.$$

.

Proof. The fundamental theorem of calculus on $[t_0, s]$; the integrand is continuous there, being a composition of continuous maps. □

Theorem 49 (Comparison lemma)**.** *Khalil, Lemma 3.4. Let f be jointly continuous and L-Lipschitz in its state argument on $[t_0, t_1]$, and suppose*

1. *u is continuous on $[t_0, t_1]$ and solves $\dot{u} = f(t, u)$ with $u(t_0) = u_0$;*
2. *v is continuous on $[t_0, t_1]$ with $v(t_0) \le u_0$, satisfies the Dini subsolution inequality $D^+v(t) \le f(t, v(t))$, and has forward difference quotients bounded above at each t;*
3. *for every $\lambda > 0$ the perturbed equation $\dot{z} = f(t, z) + \lambda$, $z(t_0) = u_0$, has a solution on $[t_0, t_1]$.*

Then $v(t) \le u(t)$ for all $t \in [t_0, t_1]$.

The perturbed solutions in (3) are taken as a hypothesis for generality, so that callers with solutions already in hand need not re-derive them. Under the hypotheses already listed they always exist: $f + \lambda$ is Lipschitz exactly when f is, so theorem 45 discharges (3).

Proof. Fix $\lambda > 0$ and let z_λ be as in hypothesis (3).

Claim: $v \le z_\lambda$ on $[t_0, t_1]$. Otherwise $v(t_{\mathrm{bad}}) > z_\lambda(t_{\mathrm{bad}})$ for some t_{bad}, and since $v(t_0) \le z_\lambda(t_0)$ the continuous difference $v - z_\lambda$ has a last zero $a < t_{\mathrm{bad}}$, with $v > z_\lambda$ on $(a, t_{\mathrm{bad}}]$. Being below v immediately to the right of a forces $\dot{z}_\lambda(a) \le D^+v(a)$, so

$$f(a, z_\lambda(a)) + \lambda \;\le\; D^+v(a) \;\le\; f(a, v(a)) = f(a, z_\lambda(a)),$$

contradicting $\lambda > 0$.

Now let $\varepsilon > 0$ and choose λ small enough that theorem 47 — which applies because f is Lipschitz in its state argument, and which treats λ as a perturbation of size λ — gives $\|u - z_\lambda\| \le \varepsilon/2$ on $[t_0, t_1]$. Then $v \le z_\lambda \le u + \varepsilon$. As ε was arbitrary, $v \le u$. □

2.4 Lie derivatives

The rate of change of a function $V(t, x)$ along the integral curves of a time-dependent vector field, with no differential-equation content beyond the chain rule: what every Lyapunov, Chetaev and invariance argument differentiates.

Definition 50 (Lie derivative along a vector field)**.** For $V : \mathbb{R} \times E \to \mathbb{R}$ and a time-dependent vector field $v : \mathbb{R} \times E \to E$, the *Lie derivative of V along v* is

$$L_v V(t, x) = \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x)\, v(t, x),$$

the derivative of V at (t, x) in the direction $(1, v(t, x))$: the Lie derivative of V, as a function on $\mathbb{R} \times E$, along the autonomous field $(t, x) \mapsto (1, v(t, x))$.

Lemma 51 (Lie derivative of a time-invariant function)**.** *If V does not depend on t and is differentiable at x, then $L_v V(t,x) = DV(x)\, v(t,x)$.*

Proof. □

Lemma 52 (Lie derivative of $V(x) + ct$)**.** *If V does not depend on t and is differentiable at x, then the Lie derivative of $(t,x) \mapsto V(x) + ct$ along v is $DV(x)\, v(t,x) + c$.*

Proof. □

Lemma 53 (Chain rule along an integral curve)**.** *Let V be differentiable and γ an integral curve of v on I. Then for $t \in I$, within I,*

$$\frac{d}{dt} V(t, \gamma(t)) = L_v V(t, \gamma(t))$$

(definition 50).

Proof. □

Lemma 54 (Nonpositive Lie derivative along an integral curve)**.** *Let V be differentiable and γ an integral curve of v on an interval s with $L_v V(t, \gamma(t)) \le 0$ at every interior point t of s. Then $t \mapsto V(t, \gamma(t))$ is nonincreasing on s.*

Proof. The mean value theorem, with the derivative given by lemma 53. □

Lemma 55 (Exponential bound from a linear Lie-derivative bound)**.** *Let V be differentiable, γ an integral curve of v on $[t_0, t_1]$, and $c \in \mathbb{R}$ with $L_v V(t, \gamma(t)) \le -c\, V(t, \gamma(t))$ for $t \in (t_0, t_1)$. Then $V(t, \gamma(t)) \le e^{-c(t-t_0)}\, V(t_0, \gamma(t_0))$ on $[t_0, t_1]$.*

Proof. The derivative of $e^{ct} V(t, \gamma(t))$ is $e^{ct}(cV + L_v V) \le 0$ (lemma 53), so it does not increase. □

Lemma 56 (Continuity of the Lie derivative)**.** *If V is C^1 and v is continuous, then $(t,x) \mapsto L_v V(t,x)$ is continuous.*

Proof. □

2.5 ω-limit sets

The points a curve or sequence returns arbitrarily close to, arbitrarily late: its cluster points at infinity, which is the ω-limit of topological dynamics for a single orbit. These are the topological facts behind LaSalle's invariance principle, in continuous and in discrete time alike: a continuous function converging along the orbit is constant on its ω-limit set, an orbit trapped in a closed set has its ω-limit set there, and an orbit trapped in a compact set converges to its ω-limit set.

Definition 57 (ω-limit set)**.** The *ω-limit set* of $\varphi : \tau \to X$, for τ a preordered index set such as $\mathbb{R}$ or $\mathbb{N}$, is the set of cluster points of φ as $t \to \infty$:

$$\omega(\varphi) = \bigcap_T \overline{\{\varphi(t) : t \ge T\}},$$

the points y such that $\varphi(t)$ is frequently in every neighbourhood of y as $t \to \infty$.

Lemma 58 (Membership in the ω-limit set)**.** *$y \in \omega(\varphi)$ (definition 57) if and only if for every neighbourhood U of y there are arbitrarily large t with $\varphi(t) \in U$.*

Proof. □

Lemma 59 (The ω-limit set as Mathlib's ω-limit)**.** *definition 57 is the ω-limit, in the sense of topological dynamics, of the one-member family $\{\varphi\}$.*

Proof. □

Lemma 60 (A limit of $g\circ\varphi$ is the value of g on $\omega(\varphi)$)**.** *Let $g : X \to Y$ be continuous, Y Hausdorff, and $g(\varphi(t)) \to L$ as $t \to \infty$. Then $g(y) = L$ for every $y \in \omega(\varphi)$ (definition 57).*

Proof. $g(y)$ is a cluster point of $g\circ\varphi$, which converges to L; in a Hausdorff space the two coincide. □

Lemma 61 (The ω-limit set of a curve trapped in a closed set)**.** *If Ω is closed and $\varphi(t) \in \Omega$ for all large t, then $\omega(\varphi) \subseteq \Omega$.*

Proof. A point outside Ω has the open neighbourhood $X \setminus \Omega$, which φ eventually avoids. □

Lemma 62 (Convergence to a set containing the ω-limit set)**.** *If $\varphi(t)$ lies in a compact set for all large t and $\omega(\varphi) \subseteq M$, then $\varphi(t) \to M$: for every open $W \supseteq M$, $\varphi(t) \in W$ for all large t.*

Proof. Mathlib's `eventually_mapsTo_of_isCompact_absorbing_of_isOpen_of_omegaLimit_subset` (lemma 59): a family eventually absorbed by a compact set is eventually inside every open set containing its ω-limit set. □

Chapter 3

Systems

The most general systems the library considers are $\dot{x} = F(t, x, u)$ in continuous time and $x^+ = F(k, x, u)$ in discrete time, each with an output $y = h(t, x, u)$. As for the linear systems of the next chapter, there is one object per time domain: the data has the same shape, but a continuous trajectory is an integral curve and a discrete one is a recursion.

Special cases are constructors on the general object, each with a predicate recognising it: *time invariance* drops t, and *autonomy* drops u. Autonomy is used here in the control-theory sense, *without input*; much of the dynamical-systems literature uses the word for time invariance instead.

3.1 Continuous time

Definition 63 (Continuous-time system)**.** A *continuous-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{R} \times E \times U \to E, \qquad h : \mathbb{R} \times E \times U \to Y,$$

to be read as $\dot{x} = F(t, x, u)$, $y = h(t, x, u)$. The structure is only the data.

Definition 64 (Time-invariant continuous-time system)**.** The continuous-time system $\dot{x} = f(x, u)$, $y = g(x, u)$, whose equations do not depend on time.

Definition 65 (Time invariance, continuous-time system)**.** A continuous-time system is *time-invariant* when $F(t, \cdot, \cdot)$ and $h(t, \cdot, \cdot)$ are the same for every t.

Lemma 66 (Time-invariant continuous-time systems are time-invariant)**.** *The system of definition 64 is time-invariant (definition 65).*

Proof. □

Lemma 67 (Time-invariant continuous-time systems come from time-free data)**.** *A time-invariant continuous-time system (definition 65) is* timeInvariant *f g for some f and g. The converse of lemma 66.*

Proof. Read f and g off at $t = 0$; time invariance says every other time agrees with it. □

Definition 68 (Autonomous continuous-time system)**.** The continuous-time system $\dot{x} = f(t, x)$, $y = g(t, x)$, which has no input: its input space is the one-point type.

Definition 69 (Autonomy, continuous-time system)**.** A continuous-time system is *autonomous* when $F(t, x, u)$ and $h(t, x, u)$ do not depend on u.

Lemma 70 (Autonomous continuous-time systems are autonomous)**.** *The system of definition 68 is autonomous (definition 69).*

Proof. □

Lemma 71 (Autonomous continuous-time systems come from input-free data)**.** *If a continuous-time system is autonomous (definition 69) and its input space is nonempty, there are f and g with $F(t, x, u) = f(t, x)$ and $h(t, x, u) = g(t, x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 72 (The unforced system)**.** The autonomous system (definition 68) $\dot{x} = F(t, x, 0)$, $y = h(t, x, 0)$ obtained by holding the input at 0. Stability of a system with input means stability of this system.

Lemma 73 (Trajectories of the unforced system)**.** *A map is a trajectory of the unforced system (definition 72) on I exactly when it is a trajectory of s under the zero input on I.*

Proof. □

Well-posedness is kept apart from stability. The stability notions quantify over every solution, so they need no uniqueness; forward completeness enters only where a limit as $t \to \infty$ is taken. Theorems that need either property assume it.

Definition 74 (Forward completeness from a set)**.** A continuous-time system is *forward complete* from a set S at time t_0 under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 81) with $x(t_0) \in S$ agrees on that segment with a trajectory on $[t_0, \infty)$.

Definition 75 (Forward completeness)**.** A continuous-time system is *forward complete* under the input u when it is forward complete (definition 74) from every state at every initial time.

Definition 76 (Uniqueness of solutions)**.** A continuous-time system has *unique solutions* under the input u when any two trajectories on a segment $[t_0, t_1]$ (definition 81) that agree at t_0 agree on all of $[t_0, t_1]$.

Theorem 77 (Lipschitz systems have unique solutions)**.** *If $x \mapsto F(t, x, u(t))$ is K-Lipschitz for every t, the system has unique solutions under u (definition 76).*

Proof. Grönwall's inequality, in the form of Mathlib's `ODE_solution_unique_of_mem_Icc_right`: a derivative within $[t_0, t_1]$ at a point of $[t_0, t_1)$ is a right derivative there. □

Theorem 78 (Globally Lipschitz systems are forward complete)**.** *If $(t, x) \mapsto F(t, x, u(t))$ is continuous and K-Lipschitz in x uniformly in t, the system is forward complete under u (definition 75).*

Proof. From the endpoint $x(t_0)$ of a segment, Picard–Lindelöf gives a solution on $[t_0, \infty)$ (theorem 44), which agrees with the segment by uniqueness (theorem 77). □

Definition 79 (State equation of a continuous-time system)**.** Under an input signal $u : \mathbb{R} \to U$, the right-hand side of the state equation is the time-dependent vector field $(t, x) \mapsto F(t, x, u(t))$.

Definition 80 (Output equation of a continuous-time system)**.** Under an input signal u, the output equation is $(t, x) \mapsto h(t, x, u(t))$. An instantaneous relation: no derivative appears.

Definition 81 (Trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when

$$\dot{x}(t) = F(t, x(t), u(t)), \qquad \forall\, t \in I,$$

the derivative being taken within I. Stated on a set: a map can satisfy the equation on an interval without being defined for all time.

Definition 82 (Trajectory for all time of a continuous-time system)**.** A trajectory (definition 81) defined on all of $\mathbb{R}$.

Lemma 83 (Trajectories of a continuous-time system on the whole line)**.** *definition 81 at $I = \mathbb{R}$ is definition 82.*

Proof. □

Lemma 84 (Restriction of a trajectory of a continuous-time system)**.** *A trajectory for all time is a trajectory on every set.*

Proof. □

A set is forward invariant when trajectories that start in it stay in it.

Definition 85 (Forward invariant set)**.** A set $S \subseteq E$ is *forward invariant* (or positively invariant) for s under the input u when every trajectory on a segment $[t_0, t_1]$ (definition 81) that starts in S stays in S:

$$x(t_0) \in S \;\Rightarrow\; x(t) \in S \quad \forall\, t \in [t_0, t_1].$$

The derivative of a function along the system is the rate at which it changes along trajectories. It takes the input value as an argument: stability uses the input-free system, input-to-state notions a general input.

Definition 86 (Derivative along the system)**.** For $V : \mathbb{R} \times E \to \mathbb{R}$ and an input value $u \in U$, the *derivative of V along the system $\dot{x} = F(t, x, u)$* is

$$\dot{V}(t, x, u) = \frac{\partial V}{\partial t}(t, x) + \frac{\partial V}{\partial x}(t, x)\, F(t, x, u),$$

the Lie derivative of V along the field $F(\cdot, \cdot, u)$ (definition 50).

The derivative form above excludes the solutions produced by discontinuous inputs, which have corners where the input jumps. The integral form admits them, and is the one input-driven notions such as input-to-state stability are stated with; for continuous right-hand sides the two agree.

Definition 87 (Carathéodory trajectory of a continuous-time system)**.** A map $x : \mathbb{R} \to E$, with E a Banach space, is a *Carathéodory trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when, for all $a \leq b$ in I, the map $\tau \mapsto F(\tau, x(\tau), u(\tau))$ is integrable on $[a, b]$ and

$$x(b) - x(a) = \int_a^b F(\tau, x(\tau), u(\tau))\, \mathrm{d}\tau.$$

Unlike definition 81 it does not require x to be differentiable, so it admits the solutions produced by discontinuous or merely measurable inputs, which input-to-state stability quantifies over.

Definition 88 (Carathéodory trajectory for all time)**.** A Carathéodory trajectory (definition 87) on all of $\mathbb{R}$.

Lemma 89 (Restriction of a Carathéodory trajectory)**.** *A Carathéodory trajectory on I is one on every subset of I; in particular one for all time is one on every set.*

Proof. □

Theorem 90 (Carathéodory and classical trajectories agree on a segment)**.** *Let $t_0 \leq t_1$ and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on $[t_0, t_1]$. Then x is a Carathéodory trajectory on $[t_0, t_1]$ (definition 87) if and only if it is a trajectory there (definition 81).*

Proof. Both are equivalent to the integral equation anchored at t_0 (definition 29), by lemma 34. The anchored equation gives the equation between any $a \leq b$ by subtracting its values at b and a. □

Theorem 91 (Carathéodory and classical trajectories agree on an interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval (order-connected) and let $\tau \mapsto F(\tau, x(\tau), u(\tau))$ be continuous on I. Then x is a Carathéodory trajectory on I (definition 87) if and only if it is a trajectory there (definition 81).*

Order-connectedness cannot be dropped: on $\{0\} \cup [1, 2]$ the integral form relates $x(1)$ to $x(0)$, the derivative form does not.

Proof. Both notions reduce to closed segments inside I, where they agree by theorem 90. For the integral form, the equation between $a \leq b$ only involves $[a, b] \subseteq I$. For the derivative form, at $t \in I$ choose $a \leq t \leq b$ in I, taking $a < t$ if I has a point left of t and $a = t$ otherwise, and likewise for b. Then $[a, b] \subseteq I$ agrees with I near t, so a derivative within $[a, b]$ at t is one within I. □

3.2 Discrete time

Definition 92 (Discrete-time system)**.** A *discrete-time system* with state space E, input space U and output space Y is the data of two maps

$$F : \mathbb{N} \times E \times U \to E, \qquad h : \mathbb{N} \times E \times U \to Y,$$

to be read as $x^+ = F(k, x, u)$, $y = h(k, x, u)$. The same data as definition 63 with time in $\mathbb{N}$; what differs is the reading, a shift rather than a derivative.

Definition 93 (Time-invariant discrete-time system)**.** The discrete-time system $x^+ = f(x, u)$, $y = g(x, u)$, whose equations do not depend on time.

Definition 94 (Time invariance, discrete-time system)**.** A discrete-time system is *time-invariant* when $F(k,\cdot,\cdot)$ and $h(k,\cdot,\cdot)$ are the same for every k.

Lemma 95 (Time-invariant discrete-time systems are time-invariant)**.** *The system of definition 93 is time-invariant (definition 94).*

Proof. □

Lemma 96 (Time-invariant discrete-time systems come from time-free data)**.** *A time-invariant discrete-time system (definition 94) is* timeInvariant *f g for some f and g. The converse of lemma 95.*

Proof. Read f and g off at $k = 0$; time invariance says every other time agrees with it. □

Definition 97 (Autonomous discrete-time system)**.** The discrete-time system $x^+ = f(k,x)$, $y = g(k,x)$, which has no input: its input space is the one-point type.

Definition 98 (Autonomy, discrete-time system)**.** A discrete-time system is *autonomous* when $F(k,x,u)$ and $h(k,x,u)$ do not depend on u.

Lemma 99 (Autonomous discrete-time systems are autonomous)**.** *The system of definition 97 is autonomous (definition 98).*

Proof. □

Lemma 100 (Autonomous discrete-time systems come from input-free data)**.** *If a discrete-time system is autonomous (definition 98) and its input space is nonempty, there are f and g with $F(k,x,u) = f(k,x)$ and $h(k,x,u) = g(k,x)$ for every input u.*

Proof. Fix any input u_0 and read f and g off at it. □

Definition 101 (State equation of a discrete-time system)**.** Under an input sequence $u : \mathbb{N} \to U$, the right-hand side of the state equation is the time-dependent next-state map $(k,x) \mapsto F(k,x,u(k))$. The same expression as definition 79; what differs is what a trajectory does with it, a shift rather than a derivative.

Definition 102 (Output equation of a discrete-time system)**.** Under an input sequence u, the output equation is $(k,x) \mapsto h(k,x,u(k))$.

Definition 103 (Trajectory of a discrete-time system)**.** A sequence $x : \mathbb{N} \to E$ is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when

$$x(k+1) = F(k,x(k),u(k)), \qquad \forall\, k \in I.$$

Membership constrains the step *out of* k, so $I = [k_0,\infty)$ is the forward recursion from k_0, and the horizon $[k_0,k_1]$ corresponds to $I = [k_0,k_1)$.

Definition 104 (Trajectory for all time of a discrete-time system)**.** A trajectory (definition 103) satisfying the recursion at every $k \in \mathbb{N}$.

Lemma 105 (Trajectories of a discrete-time system on the whole domain)**.** *definition 103 at $I = \mathbb{N}$ is definition 104.*

Proof. □

Lemma 106 (Restriction of a trajectory of a discrete-time system)**.** *A trajectory at every time is a trajectory on every set.*

Proof. □

Unlike continuous time, a trajectory needs no hypothesis to exist or to be unique: the recursion defines it from any initial time and state for all later times. So no discrete-time solution escapes in finite time, and nothing like forward completeness is ever needed.

Chapter 4

Linear systems

4.1 The linear system objects

Linear systems are a special case of the general systems of the Systems chapter: the continuous-time system with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$, and its discrete-time counterpart. The objects below carry only the coefficient data. Each is the input of a constructor into the general system, and the semantics — state equation, output equation, trajectories — are defined through that constructor, so every result about general systems applies to linear ones unchanged.

There are two objects per time domain: a *linear time-varying* (LTV) system with coefficient maps $A(t), B(t), C(t), D(t)$, and a *linear time-invariant* (LTI) system with constant matrices. An LTI system is not a different kind of system: it builds the LTV system with constant coefficients, and the general system it describes is the time-invariant system of the Systems chapter, by definition.

4.1.1 Continuous-time LTV systems

Definition 107 (Continuous-time linear time-varying system)**.** A *continuous-time linear time-varying (LTV) system* with states indexed by X, inputs by U, outputs by Y and coefficients in $\mathbb{K}$ is the data of four matrix-valued maps on $\mathbb{R}$,

$$A : \mathbb{R} \to \mathbb{K}^{X\times X}, \quad B : \mathbb{R} \to \mathbb{K}^{X\times U}, \quad C : \mathbb{R} \to \mathbb{K}^{Y\times X}, \quad D : \mathbb{R} \to \mathbb{K}^{Y\times U},$$

describing the continuous-time system (definition 63) $\dot{x} = A(t)x + B(t)u$, $y = C(t)x + D(t)u$ (definition 108).

Definition 108 (The system of an LTV system)**.** The continuous-time system (definition 63) with $F(t,x,u) = A(t)x + B(t)u$ and $h(t,x,u) = C(t)x + D(t)u$.

Definition 109 (State equation of an LTV system)**.** The right-hand side of the state equation, $(t,z) \mapsto A(t)\,z + B(t)\,u(t)$: the vector field (definition 79) of definition 108.

Lemma 110 (The unforced state equation of an LTV system)**.** *At zero input the state equation (definition 109) reduces to* $(t,z) \mapsto A(t)\,z$.

Proof. □

Lemma 111 (The state equation of an LTV system at the origin)**.** *At $z = 0$ the state equation (definition 109) contributes only the input term, $A(t)\,0 + B(t)u(t) = B(t)u(t)$. With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 112 (Output equation of an LTV system)**.** The output equation, $(t, z) \mapsto C(t)\,z + D(t)\,u(t)$: the output map (definition 80) of definition 108.

Definition 113 (Trajectory of an LTV system)**.** A map x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{R}$ when it is a trajectory (definition 81) of definition 108:

$$\dot{x}(t) = A(t)\,x(t) + B(t)\,u(t), \qquad \forall\, t \in I.$$

Definition 114 (Trajectory for all time of an LTV system)**.** A trajectory (definition 82) for all time of definition 108.

Definition 115 (Carathéodory trajectory of an LTV system)**.** A Carathéodory trajectory (definition 87) of definition 108.

Definition 116 (Time invariance of an LTV system)**.** A continuous-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 117 (Time-invariant LTV systems are time-invariant systems)**.** *If an LTV system is time-invariant (definition 116), so is the system it describes (definition 65).*

Proof. □

4.1.2 Continuous-time LTI systems

Definition 118 (Continuous-time linear time-invariant system)**.** A *continuous-time linear time-invariant (LTI) system* is the data of four matrices $A \in \mathbb{K}^{X\times X}$, $B \in \mathbb{K}^{X\times U}$, $C \in \mathbb{K}^{Y\times X}$, $D \in \mathbb{K}^{Y\times U}$, describing the LTV system (definition 107) with these constant coefficients, $\dot{x} = Ax + Bu$, $y = Cx + Du$.

Definition 119 (The LTV system of an LTI system)**.** The LTV system (definition 107) whose coefficients are the constant maps $t \mapsto A$, $t \mapsto B$, $t \mapsto C$, $t \mapsto D$.

Lemma 120 (LTI systems are time-invariant LTV systems)**.** *The LTV system of an LTI system (definition 119) is time-invariant (definition 116).*

Proof. □

Lemma 121 (Time-invariant LTV systems come from LTI systems)**.** *A time-invariant LTV system (definition 116) is the LTV system of some LTI system. The converse of lemma 120.*

Proof. Read the coefficients off at $t = 0$; time invariance says every other time agrees with it. □

Definition 122 (The system of an LTI system)**.** The continuous-time system (definition 63) of the LTV system of s (definition 108, definition 119).

Lemma 123 (An LTI system is a linear time-invariant system)**.** *The system of an LTI system (definition 122) is the time-invariant system (definition 64) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

4.1.3 Discrete-time LTV systems

Definition 124 (Discrete-time linear time-varying system)**.** A *discrete-time linear time-varying (LTV) system* is the same data as definition 107 with the coefficients indexed by $\mathbb{N}$ instead of $\mathbb{R}$, describing the discrete-time system (definition 92) $x(k+1) = A(k)x(k) + B(k)u(k)$, $y(k) = C(k)x(k) + D(k)u(k)$ (definition 125).

The forward time domain $\mathbb{N}$ rather than $\mathbb{Z}$ matches the discrete solution theory, whose state transition matrix is an ordered product over $[t_0, t)$.

Definition 125 (The system of a discrete-time LTV system)**.** The discrete-time system (definition 92) with $F(k, x, u) = A(k)x + B(k)u$ and $h(k, x, u) = C(k)x + D(k)u$.

Definition 126 (State equation of a discrete-time LTV system)**.** The right-hand side of the state equation, $(k, z) \mapsto A(k)\, z + B(k)\, u(k)$: the next-state map (definition 101) of definition 125.

Lemma 127 (The unforced state equation of a discrete-time LTV system)**.** *At zero input the state equation (definition 126) reduces to $(k, z) \mapsto A(k)\, z$.*

Proof. □

Lemma 128 (The state equation of a discrete-time LTV system at the origin)**.** *At $z = 0$ the state equation (definition 126) contributes only the input term, $A(k)\, 0 + B(k)u(k) = B(k)u(k)$. With the input switched off the origin is therefore an equilibrium of the system.*

Proof. □

Definition 129 (Output equation of a discrete-time LTV system)**.** The output equation, $(k, z) \mapsto C(k)\, z + D(k)\, u(k)$: the output map (definition 102) of definition 125.

Definition 130 (Trajectory of a discrete-time LTV system)**.** A sequence x is a *trajectory* of s under the input u on a set $I \subseteq \mathbb{N}$ when it is a trajectory (definition 103) of definition 125:

$$x(k+1) = A(k)\, x(k) + B(k)\, u(k), \qquad \forall\, k \in I.$$

Definition 131 (Trajectory for all time of a discrete-time LTV system)**.** A trajectory (definition 104) for all time of definition 125.

Definition 132 (Time invariance of a discrete-time LTV system)**.** A discrete-time LTV system is *time-invariant* when each of its four coefficients takes the same value at every pair of times.

Lemma 133 (Time-invariant discrete-time LTV systems are time-invariant systems)**.** *If a discrete-time LTV system is time-invariant (definition 132), so is the system it describes (definition 94).*

Proof. □

4.1.4 Discrete-time LTI systems

Definition 134 (Discrete-time linear time-invariant system)**.** A *discrete-time linear time-invariant (LTI) system* is the data of four matrices A, B, C, D, describing the discrete-time LTV system (definition 124) with these constant coefficients, $x(k+1) = Ax(k) + Bu(k)$, $y(k) = Cx(k) + Du(k)$.

Definition 135 (The LTV system of a discrete-time LTI system)**.** The discrete-time LTV system (definition 124) whose coefficients are the constant maps $k \mapsto A$, $k \mapsto B$, $k \mapsto C$, $k \mapsto D$.

Lemma 136 (Discrete-time LTI systems are time-invariant LTV systems)**.** *The LTV system of a discrete-time LTI system (definition 135) is time-invariant (definition 132).*

Proof. □

Lemma 137 (Time-invariant discrete-time LTV systems come from LTI systems)**.** *A time-invariant discrete-time LTV system (definition 132) is the LTV system of some LTI system. The converse of lemma 136.*

Proof. Read the coefficients off at $k = 0$. □

Definition 138 (The system of a discrete-time LTI system)**.** The discrete-time system (definition 92) of the LTV system of s (definition 125, definition 135).

Lemma 139 (A discrete-time LTI system is a linear time-invariant system)**.** *The system of a discrete-time LTI system (definition 138) is the time-invariant system (definition 93) with $f(x, u) = Ax + Bu$ and $g(x, u) = Cx + Du$: the two routes from the LTI data to a general system agree.*

Proof. □

4.2 Peano–Baker series and the state transition matrix

For a genuinely time-varying state matrix $A(t)$, this section constructs the state transition matrix $\Phi(t, t_0)$ as the Peano–Baker series and proves Theorem 5.1: Φ solves the matrix ODE $\dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0)$ with $\Phi(t_0, t_0) = I$, and consequently $x(t) := \Phi(t, t_0)x_0$ solves $\dot{x} = A(t)x$, $x(t_0) = x_0$.

Standing hypotheses. Throughout this section A is continuous and satisfies $\|A(s)\| \le M$ for all s on the segment between the two times named in each statement. Both are needed: continuity to differentiate the series term by term, the bound to make it converge at all. Statements below do not repeat them.

Definition 140 (Peano–Baker term)**.** The k-th term of the Peano–Baker series for a time-varying state matrix $A(t)$:

$$P_0(t, t_0) := I, \qquad P_{k+1}(t, t_0) := \int_{t_0}^{t} A(s)\, P_k(s, t_0)\, \mathrm{d}s.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5 (Peano-Baker series).

Definition 141 (State transition matrix)**.** The *state transition matrix* $\Phi(t, t_0)$, given by the Peano–Baker series

$$\Phi(t, t_0) := \sum_{k=0}^{\infty} P_k(t, t_0).$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.1 (Peano-Baker series).

Lemma 142 (Continuity of the Peano–Baker terms)**.** *Each term $P_k(t, t_0)$ of the Peano–Baker series is continuous in t, for A continuous.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: P_0 is constant, and P_{k+1} is the primitive of a continuous integrand (definition 140), hence continuous by the fundamental theorem of calculus. □

Lemma 143 (Factorial bound on the Peano–Baker terms)**.** *If $\|A(s)\| \le M$ for all s between t_0 and t_1, then*

$$\|P_k(t,t_0)\| \;\le\; \|I\| \cdot \frac{(M|t-t_0|)^k}{k!} \qquad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Induction on k: bound the integrand $A(s)P_k(s,t_0)$ pointwise on the segment between t_0 and t, using submultiplicativity of the matrix norm and the induction hypothesis. Split on whether $t_0 \le t$ or $t \le t_0$ and integrate the majorant via the power rule, flipping the integral's orientation (`intervalIntegral.integral_symm`) in the backward case. □

Lemma 144 (Factorial bound on a subinterval)**.** *The bound in lemma 143, uniformized to the worst case $t = t_1$:*

$$\|P_k(t,t_0)\| \;\le\; \|I\| \cdot \frac{(M|t_1-t_0|)^k}{k!} \qquad \text{for every } t \text{ between } t_0 \text{ and } t_1.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Monotonicity of $t \mapsto (M|t-t_0|)^k$ on the segment between t_0 and t_1, combined with lemma 143. □

Lemma 145 (Summability of the Peano–Baker series)**.** *The Peano–Baker series $\sum_{k=0}^{\infty} P_k(t,t_0)$ converges absolutely, for A bounded between t_0 and t (in either order).*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Compare against the exponential series $\sum_k (M|t-t_0|)^k/k!$ via lemma 143 and the comparison test. □

Lemma 146 (Continuity of the state transition matrix)**.** *$\Phi(\cdot,t_0)$ is continuous on the segment between t_0 and t_1 (including both endpoints), for A continuous and bounded by M there.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Each term $P_k(\cdot,t_0)$ is continuous (lemma 142) with a summable, point-independent bound (lemma 144); apply the Weierstrass M-test for continuity of a series. □

Theorem 147 (The state transition matrix solves the matrix ODE)**.** *The state transition matrix solves the matrix ODE*

$$\dot{\Phi}(t,t_0) = A(t)\,\Phi(t,t_0)$$

for t strictly between t_0 and t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Differentiate the Peano–Baker series term by term, using lemma 143 as the uniform bound needed for term-by-term differentiation of a series. The $k = 0$ term is constant, so peel it off and match the shifted series' derivative against $A(t)\Phi(t,t_0)$ using linearity of the sum under left multiplication by $A(t)$. □

Theorem 148 (The state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Every term of the Peano–Baker series beyond $k = 0$ integrates over the degenerate interval $[t_0, t_0]$ and vanishes. □

Theorem 149 (Peano–Baker series: existence)**.** *Hespanha, Theorem 5.1.* $x(t) := \Phi(t, t_0)\, x_0$ *solves*

$$\dot{x}(t) = A(t)\, x(t).$$

Proof. Push theorem 147 through the fixed linear map $M \mapsto M x_0$ (continuous, since the domain is finite-dimensional). □

Theorem 150 (Matrix-valued solutions)**.** *For a constant matrix* C, $Y(t) := \Phi(t, t_0)\, C$ *solves*

$$\dot{Y}(t) = A(t)\, Y(t).$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Right-multiply theorem 147 by the constant C (`HasDerivAt.mul_const`), then reassociate. □

Lemma 151 (Integral form, vector-valued)**.** $x(t) := \Phi(t, t_0)\, x_0$ *satisfies the integral equation*

$$x(t) = x_0 + \int_{t_0}^{t} A(s)\, x(s)\, \mathrm{d}s$$

for every t *between* t_0 *and* t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. The fundamental theorem of calculus applied to theorem 149, using $\Phi(t_0, t_0) = I$ (theorem 148) to fix the initial value. □

Lemma 152 (Integral form, matrix-valued)**.** *For a constant matrix* C, $Y(t) := \Phi(t, t_0)\, C$ *satisfies the integral equation*

$$Y(t) = C + \int_{t_0}^{t} A(s)\, Y(s)\, \mathrm{d}s$$

for every t *between* t_0 *and* t_1.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. The fundamental theorem of calculus applied to theorem 150, using $\Phi(t_0, t_0) = I$ (theorem 148) to fix the initial value. □

4.2.1 Uniqueness

Theorem 153 (Peano–Baker series: uniqueness)**.** *Hespanha, Theorem 5.1. Any solution z of $\dot{x} = A(t)\,x$, $x(t_0) = x_0$, for t between t_0 and t_1, coincides with $x(t) := \Phi(t, t_0)\,x_0$.*

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value x_0, the resulting bound on $\|x(t) - z(t)\|$ collapses to 0. □

Theorem 154 (Columns of the state transition matrix)**.** *Hespanha, P5.2. For every fixed i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $\dot{x} = A(t)\,x$, $x(t_0) = e_i$, where e_i is the i-th standard basis vector.*

Proof. Restatement of theorem 153 at $x_0 := e_i$, using $\Phi(t, t_0)\,e_i = (\Phi(t, t_0))_{\cdot,i}$ (`Matrix.mulVec_single_one`). □

Theorem 155 (Uniqueness of matrix-valued solutions)**.** *Any solution Y of $\dot{Y} = A(t)\,Y$, $Y(t_0) = C$, for t between t_0 and t_1, coincides with $\Phi(t, t_0)\,C$.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5.*

Proof. Apply the continuous-dependence bound of Theorem 3.4 with zero perturbation: since both solutions share the initial value C, the resulting bound on $\|Y(t) - Z(t)\|$ collapses to 0. □

4.2.2 Semigroup, invertibility and composition

Uniqueness is what makes these available: two expressions that solve the same matrix initial value problem must agree, and each identity below is an instance of that.

Theorem 156 (Semigroup property)**.** *Hespanha, P5.3. For A continuous and bounded on $[\tau, t]$, and any $\tau \le s < t$,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

Proof. $\Phi(\cdot, \tau)$, re-anchored at s, is an integral solution of the matrix ODE on the segment between s and t with initial value $\Phi(s, \tau)$, hence coincides with $\Phi(\cdot, s)\,\Phi(s, \tau)$ there by theorem 155. □

Theorem 157 (Invertibility of the state transition matrix)**.** *Hespanha, P5.4. For A continuous and bounded on the segment between τ and t, $\Phi(t, \tau)$ is nonsingular and*

$$\Phi(t, \tau)^{-1} = \Phi(\tau, t).$$

Proof. From theorem 156, generalized to arbitrary order by re-anchoring $\Phi(\cdot, \tau)$ at t and invoking uniqueness there, $\Phi(\tau, t)\Phi(t, \tau) = \Phi(\tau, \tau) = I$, so $\Phi(t, \tau)$ and $\Phi(\tau, t)$ are mutual inverses. □

Lemma 158 (Composition on a subinterval)**.** *For every z on the segment between a and b, $\Phi(z, a) = \Phi(z, b)\,\Phi(b, a)$. Composition with the* first *argument free.*

Proof. $\Phi(\cdot, a)$, re-anchored at b, is an integral solution of $\dot{Y} = A(r)Y$ with value $\Phi(b, a)$ at b; so is $\Phi(\cdot, b)\Phi(b, a)$. They agree by theorem 155. □

Theorem 159 (Semigroup property, order-free)**.** *Hespanha, P5.3. For* any *s on the segment between τ and t,*

$$\Phi(t,s)\,\Phi(s,\tau) = \Phi(t,\tau).$$

The $\tau \le s < t$ ordering in theorem 156 matches the textbook's literal statement but is not a genuine requirement; the variation-of-constants proof needs the unordered form, where $s = t_0$ is fixed and τ ranges over the whole segment.

Proof. lemma 158 gives $\Phi(s,\tau) = \Phi(s,t)\Phi(t,\tau)$; left-multiply by $\Phi(t,s)$ and cancel using $\Phi(t,s)\Phi(s,t) = I$ (theorem 157). □

Lemma 160 (Change of base time)**.** *For every τ on the segment between t_0 and t, $\Phi(t,t_0)\,\Phi(t_0,\tau) = \Phi(t,\tau)$: composition through a fixed* base *point. This differs from theorem 159 in which of the three times is required to lie between the others — here the two fixed integration endpoints are the outer ones and τ is free, which is the shape variation of constants needs.*

Proof. theorem 159 with τ itself as the middle point gives $\Phi(t,\tau)\Phi(\tau,t_0) = \Phi(t,t_0)$; right-multiply by $\Phi(t_0,\tau)$ and cancel $\Phi(\tau,t_0)\Phi(t_0,\tau) = I$. □

Lemma 161 (Continuity in the initial time)**.** *With the first argument held fixed, $\tau \mapsto \Phi(t,\tau)$ is continuous on the segment between t and t_1 — the mirror of lemma 146, which varies the first argument. Needed wherever $\Phi(t,\cdot)$ is an integrand, as in variation of constants.*

Proof. By theorem 157, $\Phi(t,\tau) = \Phi(\tau,t)^{-1}$, so the map is $\tau \mapsto \Phi(\tau,t)$ — continuous in its first argument — followed by matrix inversion, which is continuous at every unit. □

4.2.3 Variation of constants

With a forcing term $B(t)u(t)$, the solution acquires a convolution against Φ. The difficulty is that the elapsed time appears both as the upper limit of integration and inside the integrand; the proof factors the two apart rather than differentiating Φ in its second argument.

Theorem 162 (Variation of constants: existence)**.** *Hespanha, Theorem 5.2. For continuous A, B and u, the function*

$$x(t) := \Phi(t,t_0)x_0 + \int_{t_0}^{t} \Phi(t,\tau)B(\tau)u(\tau)\,d\tau$$

satisfies $\dot{x}(t) = A(t)x(t) + B(t)u(t)$ at every interior t.

Proof. The obstacle is that t appears in both the upper limit and the integrand. Factor it out: with $w(t) := x_0 + \int_{t_0}^{t} \Phi(t_0,\tau)B(\tau)u(\tau)\,d\tau$, whose integrand does not depend on t, the fundamental theorem of calculus gives w directly, and lemma 160 shows $x(t) = \Phi(t,t_0)w(t)$. Differentiating that product and cancelling $\Phi(t,t_0)\Phi(t_0,t) = I$ leaves the claim. Φ is thus only ever integrated in its second argument, never differentiated there. □

Lemma 163 (The forced response at the initial time)**.** *The forced response (definition 165) takes the value x_0 at $t = t_0$: the forcing integral is over the degenerate interval and $\Phi(t_0,t_0) = I$. Together with theorem 162 this makes it a solution of the initial value problem, not merely of the differential equation.*

Proof. □

Theorem 164 (Variation of constants: uniqueness)**.** *Hespanha, Theorem 5.2. Any two continuous integral solutions of the forced system $\dot{x} = A(t)x + B(t)u(t)$ with the same initial value x_0 agree on the segment between t_0 and t_1.*

Proof. For fixed t the right-hand side $v \mapsto A(t)v + B(t)u(t)$ is affine in v, hence Lipschitz with constant $\max(M, 1)$ where $\|A\| \leq M$. Continuous dependence on initial conditions with zero perturbation then bounds $\|z_1(t) - z_2(t)\|$ by 0. The forcing term $B(t)u(t)$ is common to both solutions and cancels in the difference, so it never enters the Lipschitz estimate. □

4.2.4 The forced response as a trajectory

The solution theory above is stated on the coefficient maps A and B, which is where the analysis lives. The results below restate it for the system object of definition 107: the variation-of-constants formula is a *trajectory* in the sense of definition 114, which the stability theory quantifies over.

Definition 165 (Forced response)**.** The *forced response* of $\dot{x} = A(t)x + B(t)u(t)$ from the state x_0 at time t_0, given by the variation-of-constants formula

$$x(t) = \Phi(t, t_0)\, x_0 + \int_{t_0}^{t} \Phi(t, \tau)\, B(\tau)\, u(\tau)\, \mathrm{d}\tau,$$

with Φ the state transition matrix (definition 141).

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, Theorem 5.2.

Definition 166 (Homogeneous response)**.** The *homogeneous response* of $\dot{x} = A(t)x$ from the state x_0 at time t_0,

$$x(t) = \Phi(t, t_0)\, x_0,$$

with Φ the state transition matrix (definition 141). This is the forced response (definition 165) with the input switched off, and the object Lyapunov stability is stated about.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Definition 8.1.

Lemma 167 (The homogeneous response at the initial time)**.** *The homogeneous response (definition 166) takes the value x_0 at $t = t_0$, since $\Phi(t_0, t_0) = I$.*

Proof. □

Lemma 168 (Continuity of the forced response)**.** *For continuous A, B and u, the forced response (definition 165) is continuous on the closed segment between t_0 and t_1, endpoints included.*

Proof. The integrating-factor identity rewrites the response as $\Phi(z, t_0)\, w(z)$ with $w(z) = x_0 + \int_{t_0}^{z} \Phi(t_0, \tau) B(\tau) u(\tau)\, \mathrm{d}\tau$, whose integrand does not depend on z — only the limit does. Continuity of w is then the ordinary continuity of a primitive, and $\Phi(\cdot, t_0)$ is continuous by lemma 146. □

Theorem 169 (Variation of constants, integral form)**.** *The forced response (definition 165) satisfies*

$$x(t) = x_0 + \int_{t_0}^{t} \left(A(s)x(s) + B(s)u(s)\right) \mathrm{d}s$$

for every t on the closed segment between t_0 and t_1.

This is theorem 162 upgraded to include the endpoints, which the differential form cannot reach: its window has t_0 as an endpoint for every choice of t_1, so t_0 is never interior to it.

Proof. The fundamental theorem of calculus, applied to the derivative on the open interval (theorem 162) together with continuity on the closed one (lemma 168), with lemma 163 fixing the initial value. □

Theorem 170 (The forced response is a trajectory)**.** *Let s be a continuous-time linear system with continuous coefficients A and B, and let u be a continuous input. Then the forced response (definition 165) from any state x_0 at any time t_0 is a trajectory of s (definition 114) — it satisfies the state equation at every $t \in \mathbb{R}$, with a two-sided derivative.*

Restriction to an interval, in particular to the ray $[t_0, \infty)$ that the stability predicates quantify over, is lemma 84.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Theorem 5.2.*

Proof. For $t \neq t_0$, apply theorem 162 on a window having t in its interior; since the window is unordered this covers $t < t_0$ as well as $t > t_0$, and continuity of A supplies the bound on the compact segment.

At $t = t_0$ no such window exists, because the formula is anchored there and t_0 is an endpoint of every window. Instead take the integral form (theorem 169) on $[t_0, t_0 + 1]$ and on $[t_0 - 1, t_0]$; each yields a one-sided derivative at t_0 (lemma 33), and the two glue to a two-sided one. □

Lemma 171 (Zero input gives the homogeneous response)**.** *With $u = 0$ the forced response (definition 165) is the homogeneous response (definition 166): $\Phi(t, t_0)x_0 + \int_{t_0}^{t} \Phi(t, \tau)B(\tau) \cdot 0 \, \mathrm{d}\tau = \Phi(t, t_0)x_0$.*

Proof. □

Theorem 172 (The homogeneous response is a trajectory)**.** *For a continuous-time linear system with continuous coefficients, the homogeneous response (definition 166) from any state x_0 at any time t_0 is a trajectory (definition 114) under zero input.*

Definition 8.1 is stated about this map, so this is the form in which the solution theory reaches the stability theory.

Proof. theorem 170 at a system with no inputs, where the forcing integral vanishes (lemma 171). The dummy system is an artefact of the proof: only A appears in the statement, and no hypothesis on B is needed. □

Theorem 173 (Solutions are homogeneous responses)**.** *Let A be continuous and let x solve $\dot{x} = A(t)x$ on $[t_0, t_1]$. Then*

$$x(t) = \Phi(t, t_0)\, x(t_0), \qquad \forall\, t \in [t_0, t_1].$$

Proof. On $[t_0, t]$ the solution satisfies the integral equation (lemma 32), so uniqueness for the Peano–Baker solution (theorem 153) identifies it with $\Phi(\cdot, t_0)x(t_0)$ there. □

Theorem 174 (The homogeneous response is a trajectory on the forward ray)**.** *The homogeneous response (definition 166) is a trajectory of the unforced system (definition 113) on $[t_0, \infty)$.*

Proof. theorem 172, restricted to the ray by lemma 84. □

Theorem 175 (Trajectories of the unforced system are homogeneous responses)**.** *Let s have continuous A and let φ be a trajectory of the unforced system on $[t_0, t_1]$ (definition 113). Then $\varphi(t) = \Phi(t, t_0)\,\varphi(t_0)$ for every $t \in [t_0, t_1]$.*

Proof. theorem 173, after reducing the unforced state equation by lemma 110. □

Theorem 176 (Linear systems are forward complete)**.** *If A is continuous, the unforced system (definition 72) of definition 108 is forward complete (definition 75).*

Proof. A trajectory on $[t_0, t_1]$ agrees there with the homogeneous response from its initial state (theorem 175), which is a trajectory on $[t_0, \infty)$ (theorem 174). □

4.2.5 The time-invariant case: the matrix exponential

For a constant state matrix the Peano–Baker series sums to the matrix exponential, which is Hespanha's (6.1)–(6.2). The point of proving it is economy: every property of Φ established above for a time-varying A then holds for $e^{A(t-t_0)}$ by specialization, with no separate LTI theory. The two exponential facts that need no solution theory at all — the semigroup law (P6.3) and the finite-polynomial form via Cayley–Hamilton (P6.5) — are proved as statements about exp and A alone.

Lemma 177 (Peano–Baker terms for a constant matrix)**.** *For a constant state matrix A, the k-th Peano–Baker term collapses to*

$$\Phi_k(t, t_0) = \frac{(t - t_0)^k}{k!}\, A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Induction on k. The $k = 0$ term is the identity. For the step, $\Phi_{k+1}(t, t_0) = \int_{t_0}^{t} A\,\Phi_k(s, t_0)\,ds$; substituting the inductive hypothesis pulls A^{k+1} out of the integral and leaves $\int_{t_0}^{t} (s - t_0)^k ds = (t - t_0)^{k+1}/(k + 1)$, which combines with $1/k!$ to give $1/(k + 1)!$. □

Lemma 178 (The LTI state transition matrix as a series)**.** *For a constant state matrix A, the state transition matrix is the power series*

$$\Phi(t, t_0) = \sum_{k=0}^{\infty} \frac{(t - t_0)^k}{k!}\, A^k.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.1).*

Proof. Sum lemma 177 termwise; the Peano–Baker series converges for any continuous state matrix, a constant one included. □

Theorem 179 (The LTI state transition matrix is $e^{A(t-t_0)}$)**.** *For a constant state matrix A, the state transition matrix is the matrix exponential:*

$$\Phi(t, t_0) = e^{A(t-t_0)}.$$

This is the bridge that lets every property already proved for Φ in the time-varying setting — existence, uniqueness, the semigroup law, invertibility, variation of constants — be reused for LTI systems by specialization rather than reproved.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, equation (6.2).*

Proof. Both sides are power series in A: lemma 178 for the left, the defining series of exp for the right. Match them termwise. □

Theorem 180 (Uniqueness of the LTI state response)**.** *Let A be constant and let z be a continuous integral solution of $\dot{x} = Ax$, $x(t_0) = x_0$, on $[t_0, t_1]$. Then $z(t) = e^{A(t-t_0)}x_0$ for every t in that interval.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, Property P6.1.*

Proof. The time-varying uniqueness theorem applied at the constant state matrix $A(\cdot) \equiv A$, whose bound $\|A(t)\| \leq \|A\|$ is immediate, then rewritten through theorem 179. There is no new ODE content: uniqueness for LTI is uniqueness for LTV at a constant matrix. □

Theorem 181 (Columns of the LTI state transition matrix)**.** *Let A be constant. For each i, the i-th column of $e^{A(t-t_0)}$ is the unique continuous integral solution of $\dot{x} = Ax$ with $x(t_0) = e_i$, the i-th standard basis vector. Equivalently, the columns of the matrix exponential are the solutions from the standard basis.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6, Property P6.2.*

Proof. theorem 180 at $x_0 = e_i$, in the column form supplied by the time-varying column-uniqueness theorem. □

4.2.6 The forced response of a time-invariant system

For a time-invariant system the state transition matrix is the matrix exponential, so the variation-of-constants formula collapses to the closed form every textbook writes.

Lemma 182 (Forced response of a time-invariant system)**.** *For constant A and B the forced response (definition 165) is*

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\, B\, u(\tau)\, \mathrm{d}\tau.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 6.*

Proof. theorem 179 identifies Φ with the matrix exponential, applied once to the homogeneous term and once under the integral sign. □

Theorem 183 (The LTI response is a trajectory)**.** *For an LTI system (definition 118) with continuous input u, the closed form*

$$t \mapsto e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-\tau)}\, B\, u(\tau)\, \mathrm{d}\tau$$

is a trajectory (definition 114) of that system.

Proof. theorem 170 at the LTV system of the LTI system (definition 119), whose coefficient maps are constant and so continuous, rewritten by lemma 182. □

Lemma 184 (Variation of constants: output equation)**.** *(Hespanha, equation (5.8)). With x as in theorem 162, the output $y(t) = C(t)x(t) + D(t)u(t)$ splits into the homogeneous response and the forced response:*

$$y(t) = C(t)\Phi(t, t_0)x_0 + \int_{t_0}^{t} C(t)\Phi(t, \tau)B(\tau)u(\tau)\, d\tau + D(t)u(t).$$

Proof. $v \mapsto C(t)v$ is a continuous linear map and so commutes with the interval integral; the integrand is continuous by lemma 161. □

4.3 The matrix exponential and complexification

4.3.1 Complexification

Spectral arguments must happen over $\mathbb{C}$, because $\mathbb{R}$ is not algebraically closed, while the systems of interest have real coefficients. Entrywise complexification is the bridge, and these record what it commutes with. The one with content is the transfer of positive definiteness: a real positive-definite matrix remains positive definite against *complex* vectors, which is what lets a Lyapunov certificate be evaluated against an eigenvector.

Definition 185 (Complexification of a real matrix)**.** For a real matrix A, its *complexification* $A_{\mathbb{C}}$ is obtained by applying $\mathbb{R} \hookrightarrow \mathbb{C}$ to each entry.

Lemma 186 (Positive definiteness survives complexification)**.** *If a real matrix P is positive definite, so is its complexification $P_{\mathbb{C}}$, as a Hermitian matrix over $\mathbb{C}$.*

Proof. Write a complex vector as $v = a + ib$ with a, b real. Then $v^* P_{\mathbb{C}} v = a^{\mathsf{T}} P a + b^{\mathsf{T}} P b$: the imaginary part cancels because P is symmetric, and the real part is positive because $v \neq 0$ forces a or b to be nonzero. □

4.3.2 Spectral facts about the exponential

These say nothing about trajectories, initial times or state transition matrices — they are facts about exp and A alone, and are proved as such. The complexification bridge is what lets spectral arguments, which must happen over $\mathbb{C}$, yield bounds on a real exponential; P6.5 collapses the exponential series to a finite polynomial in A. The eigenpair lemma is the half of spectral mapping that is pure power-series algebra, and is what turns an eigenvalue condition on A into a statement about the growth of e^{At}.

Lemma 187 (Complexification commutes with the matrix exponential)**.** *Entrywise complexification commutes with the matrix exponential: for a real matrix A,*

$$\overline{\exp(A)} = \exp(\overline{A}),$$

where $\overline{}$ denotes applying $\mathbb{R} \hookrightarrow \mathbb{C}$ entrywise. This is the bridge used to transfer spectral facts, which live over $\mathbb{C}$, to norm bounds on a real exponential.

Proof. Entrywise complexification is a continuous ring homomorphism, and exp commutes with any such. □

Lemma 188 (The exponential acts on an eigenvector by exponentiating its eigenvalue)**.** *Let M be a square matrix and let (μ, v) be an eigenpair of M, so that $Mv = \mu v$. Then*

$$e^M v = e^\mu\, v.$$

Reference: standard power-series functional calculus for the exponential.

Proof. Induction gives $M^k v = \mu^k v$ for every k. The map $N \mapsto Nv$ is linear and continuous, so it commutes with the sum defining $e^M = \sum_k (k!)^{-1} M^k$, giving $e^M v = \sum_k (k!)^{-1} \mu^k v = e^\mu v$. □

Lemma 189 (A scalar multiple scales the eigenvalues)**.** *If $Av = \mu v$ then $(cA)v = (c\mu)v$.*

Proof. □

Lemma 190 (Powers power the eigenvalues)**.** *If $Av = \mu v$ then $A^k v = \mu^k v$ for every k.*

Proof. □

Lemma 191 (A scalar shift shifts the eigenvalues)**.** *For a square matrix A over a commutative ring and scalars α, μ, a vector v satisfies $(A + \alpha I)v = (\mu + \alpha)v$ if and only if $Av = \mu v$.*

Proof. □

Lemma 192 (The kernel form of an eigenpair)**.** *$(\mu I - A)v = 0$ if and only if $Av = \mu v$.*

Proof. □

Lemma 193 (Negative spectrum gives a contractive exponential)**.** *If every eigenpair (μ, v) of the complexification of a real matrix A, with $v \neq 0$, has $\mathrm{Re}(\mu) < 0$, then $r(e^{A_{\mathbb{C}}}) < 1$.*

Reference: standard spectral mapping for the matrix exponential.

Proof. Reverse spectral mapping: every spectral value of $e^{A_{\mathbb{C}}}$ is e^μ for an eigenvalue μ of $A_{\mathbb{C}}$, and $|e^\mu| = e^{\mathrm{Re}\,\mu} < 1$. □

Theorem 194 (Semigroup property of the matrix exponential)**.** *Hespanha, P6.3. For every $t, \tau \in \mathbb{R}$,*

$$e^{At} e^{A\tau} = e^{A(t+\tau)}.$$

Proof. tA and τA commute, being scalar multiples of the same matrix, so the exponential of the sum factors. The general $e^X e^Y = e^{X+Y}$ needs commutativity and is false without it; here it is free. □

Theorem 195 (The matrix exponential as a finite polynomial)**.** *Hespanha, P6.5. There are scalar functions $\alpha_0, \dots, \alpha_{n-1} : \mathbb{R} \to \mathbb{R}$ with*

$$e^{At} = \sum_{i=0}^{n-1} \alpha_i(t)\, A^i \qquad \text{for every } t.$$

The matrix exponential — an infinite series in A — is a finite *polynomial in A of degree less than n, at the cost of coefficients that are transcendental in t.*

Proof. Cayley–Hamilton reduces every power A^k to a combination of $A^0, \dots, A^{n-1}$, say with coefficients $\bar{a}_i(k)$; substituting into the exponential series and exchanging the finite sum over i with the infinite sum over k exhibits $\alpha_i(t) = \sum_k t^k \bar{a}_i(k)/k!$ as a witness.

The exchange needs those series to converge, which holds because the reduced coefficients grow at most geometrically, $|\bar{a}_i(k)| \le CM^k$, dominating each α_i by a scalar exponential series. That growth bound is the crux of the proof and is established by iteration, not by spectral theory: "multiply by X, reduce modulo the characteristic polynomial" is a fixed linear operation on the n-dimensional space of polynomials of degree $< n$, so one step scales the ℓ^1 size of the coefficient vector by at most a constant factor. No eigenvalues and no Jordan form appear anywhere. □

4.4 The discrete-time state transition matrix

For $x(t+1) = A(t)x(t)$ the state transition matrix is an ordered product rather than a series, and the whole theory becomes finite: every proof below is an induction, with no convergence, no differentiation and no Gronwall estimate. Two differences from the continuous case are substantive rather than merely technical. There is no standing bound on A to carry, since a finite product always converges; and Φ need not be invertible, so the semigroup property genuinely requires $\tau \le s \le t$ instead of holding in any order.

Definition 196 (Discrete state transition matrix)**.** The discrete-time *state transition matrix* $\Phi(t, t_0)$:

$$\Phi(t, t_0) := \begin{cases} A(t-1)A(t-2)\cdots A(t_0+1)A(t_0) & t > t_0, \\ I & t \le t_0. \end{cases}$$

Formally, the ordered product of A over $[t_0, t)$, which for $t \le t_0$ is the empty product I. The system runs forward only, so the $t < t_0$ branch carries no meaning; it is there to make the definition total.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 5, equation (5.11).

Lemma 197 (The discrete state transition matrix at the initial time)**.** $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Property P5.5.*

Proof. The defining product is over the empty range $[t_0, t_0)$. □

Lemma 198 (One-step recursion)**.** *For $t \ge t_0$, $\Phi(t+1, t_0) = A(t)\,\Phi(t, t_0)$. With lemma 197 this is the existence half of Hespanha's P5.5: the discrete state transition matrix solves the matrix recursion.*

Proof. Peel the leading factor $A(t)$ off the defining product. □

Theorem 199 (Discrete state transition matrix: uniqueness)**.** *Hespanha, P5.5. If $Z(t_0) = I$ and $Z(t+1) = A(t)Z(t)$ for all $t \ge t_0$, then $Z(t) = \Phi(t, t_0)$ for all $t \ge t_0$.*

Proof. Induction on t from t_0: the base case is lemma 197 and the step chains the recursion against lemma 198. No analytic content at all, unlike the continuous-time counterpart — the recursion pins each value outright. □

Theorem 200 (Uniqueness of the discrete state response)**.** *If $z(t_0) = x_0$ and $z(t+1) = A(t)z(t)$ for all $t \ge t_0$, then $z(t) = \Phi(t, t_0)x_0$ for all $t \ge t_0$: the vector form of theorem 199.*

Proof. □

Theorem 201 (Columns of the discrete state transition matrix)**.** *Hespanha, P5.6. For each i, the i-th column of $\Phi(t, t_0)$ is the unique solution of $z(t+1) = A(t)z(t)$ with $z(t_0) = e_i$, for $t \geq t_0$.*

Proof. theorem 200 at $x_0 = e_i$, using $\Phi(t, t_0)e_i = (\Phi(t, t_0))_{\cdot,i}$. □

Theorem 202 (Semigroup property, discrete)**.** *Hespanha, P5.7. For $\tau \leq s \leq t$,*

$$\Phi(t, s)\,\Phi(s, \tau) = \Phi(t, \tau).$$

Unlike the continuous-time case there is no inverse: Φ is a product of matrices that need not be nonsingular, so the ordering $\tau \leq s \leq t$ is a genuine restriction here rather than an artifact.

Proof. Induction on t from s, chaining lemma 198 at both s and τ. □

4.4.1 Discrete variation of constants

Lemma 203 (Discrete variation of constants at the initial time)**.** *The discrete variation-of-constants formula*

$$x(t) := \Phi(t, t_0)x_0 + \sum_{\tau=t_0}^{t-1} \Phi(t, \tau+1)B(\tau)u(\tau)$$

takes the value x_0 at $t = t_0$: the forcing sum is over the empty range and $\Phi(t_0, t_0) = I$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 5, Section 5.3 (discrete variation of constants).*

Proof. □

Theorem 204 (Discrete variation of constants: existence)**.** *For $t \geq t_0$, the formula of lemma 203 satisfies the forced recursion*

$$x(t+1) = A(t)\,x(t) + B(t)\,u(t).$$

Proof. Peel the top term off the sum defining $x(t+1)$ and fold $A(t)\Phi(t, \tau+1)$ into $\Phi(t+1, \tau+1)$ termwise by lemma 198, applied once more to the leading $\Phi(t, t_0)$ term. The peeled term is exactly $B(t)u(t)$, since $\Phi(t+1, t+1) = I$. □

Theorem 205 (Discrete variation of constants: uniqueness)**.** *Any two solutions of $z(t+1) = A(t)z(t) + B(t)u(t)$ with the same value at t_0 agree for all $t \geq t_0$.*

Proof. Induction on t from t_0. Unlike theorem 164, no Lipschitz condition or Gronwall estimate is needed: the recursion determines each successive value outright, so uniqueness is immediate. □

Chapter 5

Comparison functions

Comparison functions are the vocabulary of stability estimates: a class $\mathcal{K}$ function bounds how far a trajectory can go, a class $\mathcal{L}$ function how fast it decays, and a class $\mathcal{KL}$ function both at once. This chapter builds them, and the tools that turn a cruder bound into one of them.

5.1 Class $\mathcal{K}$ and $\mathcal{K}_\infty$ functions

Definition 206 (Class $\mathcal{K}$ function)**.** A *class* $\mathcal{K}$ function on $[0, a)$ is a continuous strictly increasing map $\alpha : [0, a) \to [0, b)$ with $\alpha(0) = 0$. All class K functions have an inverse $\alpha^{-1} : [0, b) \to [0, a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}$). Differences:

- Khalil writes the codomain as $[0, \infty)$; here the bound b is tracked in the type.
- The inverse is bundled as data rather than derived.

Definition 207 (Class $\mathcal{K}_\infty$ function)**.** A *class* $\mathcal{K}_\infty$ function is a continuous strictly increasing map $\alpha : [0, \infty) \to [0, \infty)$ with $\alpha(0) = 0$ and $\alpha(r) \to \infty$ as $r \to \infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.2 (class $\mathcal{K}_\infty$). The inverse is bundled as data rather than derived.

5.1.1 Building class $\mathcal{K}$ functions

A recognition criterion and the closure properties. The restatements of the definition itself (strict monotonicity as an iff, the two inverse equations, and so on) are left to the Lean API; they say nothing the definition above does not.

Definition 208 (Class $\mathcal{K}$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$ and $f(a) = b$ and be continuous and strictly increasing on $[0, a]$, with $a, b > 0$. Then f is a class $\mathcal{K}$.

Definition 209 (Restriction of a class $\mathcal{K}$ function)**.** If α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and $0 < c < a$, then α restricted to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, \alpha(c))$.

Definition 210 (Restriction to a prescribed codomain)**.** definition 209 with the codomain bound named: if $\alpha(c) = e$, the restriction of α to $[0, c)$ is class $\mathcal{K}$ on $[0, c) \to [0, e)$.

Definition 211 (Inverse of a class $\mathcal{K}$ function)**.** Class $\mathcal{K}$ is closed under inversion. The inverse of a class K function is also class K.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, first bullet. Khalil's domain $[0, \alpha(a))$ is the codomain bound b carried by the type.

Definition 212 (Composition of class $\mathcal{K}$ functions)**.** Class $\mathcal{K}$ is closed under composition: if α is class $\mathcal{K}$ on $[0, a) \to [0, b)$ and β is class $\mathcal{K}$ on $[0, b) \to [0, c)$, then $\beta \circ \alpha$ is class $\mathcal{K}$ on $[0, a) \to [0, c)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, third bullet.

5.1.2 Building class $\mathcal{K}_\infty$ functions

The same recognition criterion and closure properties on the unbounded domain, together with the closure properties that need the global domain (sum, positive scaling, minimum), the power functions as the standard examples, and the restriction back to class $\mathcal{K}$.

Definition 213 (Class $\mathcal{K}_\infty$ from a strictly increasing function)**.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$, be continuous and strictly increasing on $[0, \infty)$, and have $f(r) \to \infty$ as $r \to \infty$. Then f is a class $\mathcal{K}_\infty$ function, its inverse being the inverse of f on $[0, \infty)$.

Definition 214 (Inverse of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under inversion: if α is class $\mathcal{K}_\infty$, so is α^{-1}.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, second bullet.

Definition 215 (Composition of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under composition: if α and β are class $\mathcal{K}_\infty$, so is $\beta \circ \alpha$, with inverse $\alpha^{-1} \circ \beta^{-1}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fourth bullet.

Definition 216 (Sum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise addition: if α and β are class $\mathcal{K}_\infty$, so is $\alpha + \beta$.

Definition 217 (Positive multiple of a class $\mathcal{K}_\infty$ function)**.** Class $\mathcal{K}_\infty$ is closed under multiplication by a positive constant: if α is class $\mathcal{K}_\infty$ and $c > 0$, so is $c\alpha$.

Definition 218 (Minimum of class $\mathcal{K}_\infty$ functions)**.** Class $\mathcal{K}_\infty$ is closed under pointwise minimum: if α and β are class $\mathcal{K}_\infty$, so is $\min(\alpha, \beta)$.

Definition 219 (Powers of a class $\mathcal{K}_\infty$ function)**.** For every $p > 0$, the map $r \mapsto r^p$ is class $\mathcal{K}_\infty$.

Definition 220 (A class $\mathcal{K}_\infty$ function restricts to class $\mathcal{K}$)**.** Every class $\mathcal{K}_\infty$ function α restricts, for each $a > 0$, to a class $\mathcal{K}$ function on $[0, a) \to [0, \alpha(a))$.

Lemma 221 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let $\omega : \mathbb{R} \to \mathbb{R}$ satisfy $\omega(0) = 0$, be monotone on $[0, \infty)$, and be continuous at 0 from the right. Then there is a class $\mathcal{K}_\infty$ function α with $\omega(r) \le \alpha(r)$ for all $r \ge 0$.*

Proof. lemma 249 supplies a continuous, strictly increasing f with $f(0) = 0$, $f(r) \to \infty$ and $\omega \le f$ on $[0, \infty)$; by definition 213 that f is class $\mathcal{K}_\infty$. □

5.2 Inverses of antitone functions

A function W that is continuous and strictly antitone on $(0,\infty)$, blows up at 0^+ and vanishes at $+\infty$ surjects onto $(0,\infty)$, and its right inverse has the same shape. This is the mechanism behind the class $\mathcal{KL}$ construction of the next section: W is a mean-value bound and its inverse supplies the decay rate.

Lemma 222 (Surjectivity onto $(0,\infty)$ from the two boundary limits)**.** *Let $W : \mathbb{R} \to \mathbb{R}$ be continuous on $(0,\infty)$ with $W(\eta) \to +\infty$ as $\eta \to 0^+$ and $W(\eta) \to 0$ as $\eta \to +\infty$. Then for every $s > 0$ there exists $c > 0$ with $W(c) = s$.*

Proof. □

Lemma 223 (The right inverse inverts)**.** *Under the surjectivity conditions of lemma 222, $W\,(\mathrm{invFunOn}\,W\,(0,\infty)\,s) = s$ for all $s > 0$.*

Proof. □

Lemma 224 (Positivity of the right inverse)**.** *Under the surjectivity conditions of lemma 222,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s > 0$ *for all $s > 0$.*

Proof. □

Lemma 225 (The inverse of a strictly antitone function is strictly antitone)**.** *If W is strictly antitone on $(0,\infty)$, then* $\mathrm{invFunOn}\,W\,(0,\infty)$ *is strictly antitone on $(0,\infty)$.*

Proof. □

Lemma 226 (The inverse vanishes at $+\infty$)**.** *Under the conditions of lemma 222 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s \to 0$ *as $s \to +\infty$.*

Proof. □

Lemma 227 (Continuity of the inverse)**.** *Under the conditions of lemma 222 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)$ *is continuous on $(0,\infty)$.*

Proof. □

Lemma 228 (The inverse blows up at 0^+)**.** *Under the conditions of lemma 222 and with W strictly antitone,* $\mathrm{invFunOn}\,W\,(0,\infty)\,s \to +\infty$ *as $s \to 0^+$.*

Proof. □

5.3 Class $\mathcal{KL}$ functions and the Osgood construction

Definition 229 (Class $\mathcal{L}$ function)**.** A *class $\mathcal{L}$* function on $[0,\infty)$ is continuous, strictly positive, and strictly decreasing, and satisfies $\sigma(s) \to 0$ as $s \to \infty$. It carries the time decay of an asymptotic stability estimate: the product of a class $\mathcal{K}$ function of the initial deviation with a class $\mathcal{L}$ function of elapsed time is a class $\mathcal{KL}$ bound.

Definition 230 (Singular class $\mathcal{L}$ function)**.** A *singular class* $\mathcal{L}$ function on $(0,\infty)$ is continuous, strictly positive and antitone, with $U(s)\to 0$ as $s\to\infty$ and $U(s)\to\infty$ as $s\to 0^+$. It differs from class $\mathcal{L}$ in two ways: the domain excludes 0, where the function is instead required to blow up, and the decrease is only required to be weak. It arises as the inverse of the sliding-window function in the class $\mathcal{KL}$ characterization of asymptotic stability.

Definition 231 (Class $\mathcal{KL}$ function)**.** A *class* $\mathcal{KL}$ function on $[0,a)\times[0,\infty)$ is continuous, class $\mathcal{K}$ in the first argument, and for each fixed $r>0$ is strictly decreasing and tends to 0 as $s\to\infty$. It arises as the bound $\|x(t)\|\le\beta(\|x_0\|,t)$ in asymptotic stability estimates.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3. Difference: Khalil says $\beta(\cdot,s)$ is class $\mathcal{K}$; here that is spelled out as separate fields, so no class $\mathcal{K}$ instance is required for each s.

Definition 232 (Global class $\mathcal{KL}$ function)**.** A *global class* $\mathcal{KL}$ function is a map $\beta:[0,\infty)\times[0,\infty)\to\mathbb{R}$ that is class $\mathcal{K}_\infty$ in the first argument (continuous, strictly increasing, zero at zero, radially unbounded) and, for each fixed $r\ge 0$, is antitone and tends to 0 as $s\to\infty$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.3 with $a=\infty$.

5.3.1 Building class $\mathcal{KL}$ functions

Two constructions and the two composition rules, in the bounded and the global setting. The product $\alpha(r)\gamma(s)$ of a class $\mathcal{K}$ and a class $\mathcal{L}$ function is the easy one; the Sontag-style cap is the one that does real work.

Definition 233 (Class $\mathcal{KL}$ from a class $\mathcal{K}$ times a class $\mathcal{L}$)**.** If α is class $\mathcal{K}$ on $[0,a)$ and γ is class $\mathcal{L}$, then $\beta(r,s):=\alpha(r)\gamma(s)$ is class $\mathcal{KL}$ on $[0,a)$.

The cap is built in the form $\min(\alpha(r),\sqrt{\alpha(r)\,U(s)})$, so the decay of that particular expression is recorded first.

Lemma 234 (Decay of a geometric-mean cap)**.** *Let $c\ge 0$ and let $U:\mathbb{R}\to\mathbb{R}$ satisfy $U(s)\to 0$ as $s\to+\infty$. Then $\min(c,\sqrt{c\,U(s)})\to 0$ as $s\to+\infty$.*

Proof. □

Theorem 235 (Class $\mathcal{KL}$ from a singular class $\mathcal{L}$ cap)**.** *Let α be class $\mathcal{K}$ on $[0,a)$ and let U be a singular class $\mathcal{L}$ function (positive, antitone, $U(s)\to 0$ as $s\to\infty$ and $U(s)\to\infty$ as $s\to 0^+$). Then*

$$\beta(r,s):=\begin{cases}\alpha(r) & s=0,\\ \min(\alpha(r),\sqrt{\alpha(r)U(s)}) & s>0,\end{cases}$$

is class $\mathcal{KL}$ on $[0,a)$.

Proof. Every field but continuity is immediate from the corresponding property of α and U, the decay in s being lemma 234. Continuity is checked in two regimes. For $s>0$ the min branch is a composition of continuous maps. At $(r,0)$, pick a buffer radius $r_1\in(r,a)$; since $U(s)\to\infty$ as $s\to 0^+$, on a neighbourhood we have $\alpha(r_1)<U(s)$, hence $\alpha(r')\le U(s)$ and so $\alpha(r')\le\sqrt{\alpha(r')U(s)}$ — the minimum is locally just α, which is continuous. □

Definition 236 (Post-composition with a class $\mathcal{K}$ function)**.** Let β be class $\mathcal{KL}$ on $[0,a)$ and α be class $\mathcal{K}$ on $[0,b)$, and suppose $\beta(r,s) < b$ for every $r \in [0,a)$ and $s \geq 0$. Then $(r,s) \mapsto \alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1). The range hypothesis is additional: α_1 is defined only on $[0,b)$, so β's values must land there, which Khalil leaves implicit.

Definition 237 (Post-composition with a class $\mathcal{K}_\infty$ function)**.** If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \alpha(\beta(r,s))$ is class $\mathcal{KL}$ on $[0,a)$. This is definition 236 with the range hypothesis discharged by α being defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet, with α_1 of class $\mathcal{K}_\infty$; no range hypothesis is then needed.

Definition 238 (Pre-composition in the radius argument)**.** If β is class $\mathcal{KL}$ on $[0,b)$ and α is class $\mathcal{K}$ on $[0,a) \to [0,b)$, then $(r,s) \mapsto \beta(\alpha(r),s)$ is class $\mathcal{KL}$ on $[0,a)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2).

Definition 239 (Restriction of a class $\mathcal{KL}$ function)**.** If β is class $\mathcal{KL}$ on $[0,a) \times [0,\infty)$ and $0 < a' \leq a$, then β is class $\mathcal{KL}$ on $[0,a') \times [0,\infty)$.

Definition 240 (Global class $\mathcal{KL}$ from a product)**.** If α is class $\mathcal{K}_\infty$ and γ is class $\mathcal{L}$, then $\beta(r,s) := \alpha(r)\gamma(s)$ is global class $\mathcal{KL}$.

Definition 241 (Post-composition, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \alpha(\beta(r,s))$ is global class $\mathcal{KL}$. No range hypothesis is needed, since α is defined on all of $[0,\infty)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the outer α_1), in the global case $a = \infty$.

Definition 242 (Pre-composition in the radius argument, global case)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \beta(\alpha(r),s)$ is global class $\mathcal{KL}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Lemma 4.2, fifth bullet (the inner α_2), in the global case $a = \infty$.

Theorem 243 (Capping a class $\mathcal{KL}$ function by a class $\mathcal{K}$ function)**.** *If β is class $\mathcal{KL}$ on $[0,a)$ and α is class $\mathcal{K}$ on $[0,a)$, then $(r,s) \mapsto \min(\beta(r,s),\alpha(r))$ is class $\mathcal{KL}$ on $[0,a)$.*

Capping a class $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class. Its use is to recover a range condition: a bound obtained only as a majorant — from a smoothing construction, say — carries no upper control, and capping restores it without disturbing the bound, since the minimum still dominates whatever β dominated.

Proof. Each field is inherited. The minimum of two strictly increasing functions of r is strictly increasing; α does not depend on s, so antitonicity in s survives the minimum; and $\min(\beta(r,s),\alpha(r)) \to \min(0,\alpha(r)) = 0$ as $s \to \infty$ because $\alpha \geq 0$. □

Definition 244 (Capping a global class $\mathcal{KL}$ function)**.** If β is global class $\mathcal{KL}$ and α is class $\mathcal{K}_\infty$, then $(r,s) \mapsto \min(\beta(r,s),\alpha(r))$ is global class $\mathcal{KL}$ — capping a $\mathcal{KL}$ bound by a function of the initial condition alone preserves the class.

5.3.2 Smoothing a bound into a class $\mathcal{KL}$ function

A candidate bound that has the right qualitative behaviour — monotone in the initial deviation, decaying in time — need not itself be class $\mathcal{KL}$: it may fail continuity or strict monotonicity. These two results replace such a candidate by a genuine class $\mathcal{KL}$ function dominating it, which is how the stability characterizations produce their bounds. The first lemma below is the continuity criterion the construction rests on.

Lemma 245 (Joint continuity from separate continuity and monotonicity)**.** *Let $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be nondecreasing in its first argument, nonincreasing in its second, and continuous in each argument separately. Then f is jointly continuous.*

Proof. Near (r_0, s_0), monotonicity gives $f(r_-, s_+) \le f(r, s) \le f(r_+, s_-)$ whenever $r_- < r < r_+$ and $s_- < s < s_+$. Given $u > f(r_0, s_0)$, continuity of $f(r_0, \cdot)$ gives $s_- < s_0$ with $f(r_0, s_-) < u$, and then continuity of $f(\cdot, s_-)$ gives $r_+ > r_0$ with $f(r_+, s_-) < u$; so $f < u$ on the neighbourhood $(-\infty, r_+) \times (s_-, \infty)$. The lower bound is symmetric. □

Lemma 246 (Class $\mathcal{KL}$ majorant)**.** *Let $\psi : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$ satisfy, for all $r, s \ge 0$:*

1. *$\psi(r, s) \ge 0$;*
2. *$\psi(0, s) = 0$;*
3. *$r \mapsto \psi(r, s)$ is monotone on $[0, \infty)$;*
4. *$s \mapsto \psi(r, s)$ is antitone on $[0, \infty)$;*
5. *$\psi(r, s) \to 0$ as $s \to \infty$;*
6. *$r \mapsto \psi(r, 0)$ is continuous at 0 within $[0, \infty)$.*

Then there is a global class $\mathcal{KL}$ function β with $\psi(r, s) \le \beta(r, s)$ for all $r, s \ge 0$.

Proof. Clamp both arguments of ψ at 0 and average it over a window in each variable: first $\psi_1(r, s) = \int_0^1 \psi(r, s - v)\, dv$, over $[s-1, s]$, then $\psi_2(r, s) = \int_1^2 \psi_1(ru, s)\, du$, over $[r, 2r]$. Each window sits on the side that makes the average dominate, so $\psi \le \psi_1 \le \psi_2$, and ψ_2 is still nondecreasing in r and nonincreasing in s.

- $\psi_1(r, \cdot)$ is a difference of primitives of $\psi(r, \cdot)$, so continuous; $\psi_2(r, \cdot)$ is then continuous by dominated convergence, the integrands being bounded by $\psi(2r, 0)$.
- For $r > 0$, $\psi_2(r, s) = r^{-1} \int_r^{2r} \psi_1(\cdot, s)$ is a quotient of primitives, so continuous; at $r \le 0$, $0 \le \psi_2(r, s) \le \psi(2 \max(r, 0), 0) \to \psi(0, 0) = 0$ by continuity of $\psi(\cdot, 0)$ at 0.

By lemma 245, ψ_2 is jointly continuous. Also $\psi_2(r, s) \le \psi(2r, s - 1) \to 0$. Finally $\beta(r, s) = \psi_2(r, s) + re^{-s}$ is class $\mathcal{KL}$, the second term making it strictly increasing in r. □

Lemma 247 (Global class $\mathcal{KL}$ from a decay family)**.** *Let α be class $\mathcal{K}_\infty$ and let U be a family of functions such that*

1. *$U(r, s) > 0$ for all $r > 0$ and $s > 0$;*

2. $U(r,\cdot)$ *is antitone on* $(0,\infty)$ *for each* $r>0$*;*

3. $U(r,s)\to 0$ *as* $s\to\infty$*, for each* $r>0$*;*

4. $r\mapsto U(r+1,s)$ *is monotone on* $[0,\infty)$ *for each* $s>0$*.*

Then there is a global class $\mathcal{KL}$ *function* β *such that*

1. $\beta(r,0)\to\infty$ *as* $r\to\infty$*;*

2. $\alpha(r)\le\beta(r,0)$ *for all* $r\ge 0$*;*

3. $\min(\alpha(r),\sqrt{\alpha(r)U(r+1,s)})\le\beta(r,s)$ *for all* $r\ge 0$ *and* $s>0$*.*

Proof. Take $\psi(r,s):=\alpha(r)$ for $s=0$ and $\min(\alpha(r),\sqrt{\alpha(r)U(r+1,s)})$ for $s>0$, the radius being shifted by 1 so that U is only ever evaluated at a strictly positive radius. Each hypothesis of lemma 246 holds for ψ: monotonicity in r from that of α and of $r\mapsto U(r+1,s)$, antitonicity in s from that of $U(r+1,\cdot)$, decay from lemma 234, and continuity at $r=0$ because $\psi(\cdot,0)=\alpha$. The resulting β dominates ψ, which gives the three conclusions; $\beta(r,0)\ge\alpha(r)\to\infty$ gives the first. No property of ψ beyond these bounds appears in the statement. □

5.4 Class $\mathcal{K}$ bounds for positive definite functions

A Lyapunov function is positive definite: a pointwise condition, awkward to combine with estimates. The theorem below converts it into monotone bounds in the single variable $\|x\|$, after which every Lyapunov argument can be run in the comparison-function calculus. It rests on two smoothing results, which replace a monotone bound by a continuous strictly increasing one.

Lemma 248 (Class $\mathcal{K}$ minorant of a positive monotone function)**.** *Let* $r>0$ *and let* $\psi:\mathbb{R}\to\mathbb{R}$ *satisfy* $\psi(0)=0$*,* $\psi(s)>0$ *for* $s\in(0,r]$*, and* ψ *nondecreasing on* $[0,r]$*. Then there are* $b>0$ *and a function* f *with* $f(0)=0$*,* $f(r)=b$*,* f *continuous and strictly increasing on* $[0,r]$*, and* $f(s)\le\psi(s)$ *for all* $s\in[0,r]$*.*

That is: a positive nondecreasing function admits a class $\mathcal{K}$ *minorant on any bounded interval. This is what turns a positive-definite* V *into the lower comparison bound of theorem 251.*

Proof. Take $f(s)=r^{-1}\int_0^s\psi$ (with the argument of ψ clamped to $[0,r]$, so that it is monotone, hence integrable, on all of $\mathbb{R}$). It is continuous as a primitive, and $f(0)=0$. For $s<s'$ in $[0,r]$, $f(s')-f(s)=r^{-1}\int_s^{s'}\psi>0$ since $\psi>0$ on (s,s'), so f is strictly increasing and $b=f(r)>0$. Finally, ψ is nondecreasing, so $\int_0^s\psi\le s\,\psi(s)\le r\,\psi(s)$, i.e. $f(s)\le\psi(s)$. □

Lemma 249 (Class $\mathcal{K}_\infty$ majorant of a monotone function)**.** *Let* $\varphi:\mathbb{R}\to\mathbb{R}$ *satisfy* $\varphi(0)=0$*, be monotone on* $[0,\infty)$*, and be continuous at* 0 *from the right. Then there is an* f *with* $f(0)=0$*, continuous and strictly increasing on* $[0,\infty)$*,* $f(s)\to\infty$ *as* $s\to\infty$*, and* $\varphi(s)\le f(s)$ *for all* $s\ge 0$*.*

That is: a nondecreasing function admits a class $\mathcal{K}_\infty$ *majorant. The unbounded domain is what distinguishes this from lemma 250, which it implies.*

Continuity at 0 *is necessary: the unit step has no continuous majorant vanishing at* 0*.*

Proof. Take $f(s) = s + g(s)$ with $g(s) = \int_1^2 \varphi(su)\,du$, the average of φ over $[s, 2s]$ (with φ clamped to $\varphi(\max(t, 0))$ so that it is monotone, hence integrable, on all of $\mathbb{R}$). Monotonicity of φ gives $\varphi(s) \le g(s) \le \varphi(2s)$ and that g is nondecreasing, so f is strictly increasing, $f \ge \varphi$, and $f(s) \ge s \to \infty$. For $s > 0$, $g(s) = s^{-1}(\Phi(2s) - \Phi(s))$ with Φ the (continuous) primitive of φ, so g is continuous there; at 0, $0 \le g(s) \le \varphi(2s) \to 0$ by continuity of φ at 0. □

Lemma 250 (Class $\mathcal{K}$ majorant on a bounded interval)**.** *Let $r > 0$ and let φ satisfy $\varphi(0) = 0$, be monotone on $[0, r]$, and be continuous at 0 from the right. Then there are $b > 0$ and an f with $f(0) = 0$, $f(r) = b$, f continuous and strictly increasing on $[0, r]$, and $\varphi(s) \le f(s)$ for all $s \in [0, r]$.*

Proof. Clamp φ to $\varphi_{\text{ext}}(s) = \varphi(\min(s, r))$, which is monotone on all of $[0, \infty)$, still vanishes at 0 and agrees with φ near 0 (so is still continuous there), and apply lemma 249. Restricting the resulting majorant to $[0, r]$ gives f; $b = f(r) > 0$ because f is strictly increasing from $f(0) = 0$. □

Theorem 251 (Class $\mathcal{K}$ sandwich bounds)**.** *Let $W : \mathbb{R}^n \to \mathbb{R}$ be positive definite (definition 25) about x_{eq} on a set containing $\overline{B}(x_{\text{eq}}, r)$. Then there are class $\mathcal{K}$ functions α_1, α_2 on $[0, r)$ with*

$$\alpha_1(\|x - x_{\text{eq}}\|) \;\le\; W(x) \;\le\; \alpha_2(\|x - x_{\text{eq}}\|) \qquad \forall\, \|x - x_{\text{eq}}\| < r.$$

The conclusion holds on the open *ball: α_1 and α_2 are class $\mathcal{K}$ on $[0, r)$, so neither is defined at $\|x - x_{\text{eq}}\| = r$.*

This is what lets a Lyapunov argument be run entirely in terms of comparison functions: positive definiteness is a pointwise condition, and this converts it into monotone bounds in the single variable $\|x - x_{\text{eq}}\|$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.3 (proved in his Appendix C.4). Differences:*

- *The bound holds on the open ball, with α_1, α_2 class $\mathcal{K}$ on $[0, r)$; Khalil states it on $[0, r]$.*
- *The radially unbounded $\mathcal{K}_\infty$ case is not derived here.*

Proof. Translate to the origin, $V(y) = W(y + x_{\text{eq}})$. For the lower bound take $\psi(s) = \inf\{V(y) : s \le \|y\| \le r\}$, the infimum of V over the annulus of inner radius s: it vanishes at 0, is positive for $s > 0$ by compactness and positive definiteness, and is nondecreasing. For the upper bound take $\phi(s) = \sup\{V(y) : \|y\| \le s\}$, similarly nondecreasing and vanishing at 0. Neither is continuous or strictly monotone, so smooth them: lemma 248 gives a class $\mathcal{K}$ minorant of ψ and lemma 250 a class $\mathcal{K}$ majorant of ϕ. Since $\psi(\|y\|) \le V(y) \le \phi(\|y\|)$ pointwise, these sandwich V. □

5.5 The decay system and the comparison bound

A class $\mathcal{K}$ decay rate produces a class $\mathcal{KL}$ bound. This is the step that turns a Lyapunov estimate — which controls only the *rate* of decrease — into an asymptotic stability statement, and it is where the comparison system $\dot{y} = -\alpha(y)$ enters: a Lyapunov function along a trajectory is a subsolution of it.

Definition 252 (The decay system of a class $\mathcal{K}$ function)**.** For α class $\mathcal{K}$ on $[0, a)$ (definition 206), the *decay system* is the scalar time-invariant system

$$\dot{y} = -\alpha(y),$$

as a continuous-time system (definition 63) with no input. Its meaningful states are $y \in [0, a)$.

Theorem 253 (The decay system has a class $\mathcal{KL}$ solution operator)**.** *Let α be class $\mathcal{K}$ on $[0,a)$ and at most linear near the origin, $\alpha(x) \le Lx$ on $(0,\text{base}]$. Then there is a class $\mathcal{KL}$ function σ (definition 231) with $\sigma(y_0,0)=y_0$ such that for every $y_0 \in [0,a)$ and every t_0, $t \mapsto \sigma(y_0, t-t_0)$ is a trajectory on $[t_0,\infty)$ (definition 81) of the decay system $\dot{y} = -\alpha(y)$ (definition 252).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.4. Differences:*

- *Khalil assumes α locally Lipschitz; here the hypothesis is the linear bound $\alpha(x) \le Lx$ near the origin, which is what the construction uses.*
- *Uniqueness of the solution is not asserted.*

Proof. **Osgood's construction.** Let $\eta(y) = -\int_{\text{base}}^{y} \mathrm{d}x/\alpha(x)$ be the time for the solution started at base to reach y. Since $\alpha > 0$ on $(0,a)$, η is continuous and strictly decreasing there; and since $\alpha(x) \le Lx$ near 0, $\eta(y) \ge (\log \text{base} - \log y)/L \to +\infty$ as $y \to 0^+$, so the origin is not reached in finite time. Setting $\sigma(r,s) := \eta^{-1}(\eta(r)+s)$, extended by $\sigma(0,s)=0$, the class $\mathcal{KL}$ conditions follow from monotonicity and continuity of η^{-1}, and differentiating $\eta(\sigma) = \eta(r)+s$ gives $\dot{\sigma} = -\alpha(\sigma)$ for $s>0$. At $s=0$ the one-sided derivative is the limit of these, $-\alpha(r)$, by continuity of σ and α. □

Lemma 254 (Lipschitz class $\mathcal{K}$ minorant)**.** *Let α be a class $\mathcal{K}$ function on $[0,a)$. Then there are $b'>0$, $L>0$ and a class $\mathcal{K}$ function $\beta : [0,a) \to [0,b')$ with $\beta(x) \le \alpha(x)$ for $x \in [0,a)$, $\beta(x) \le Lx$ for $x \ge 0$, and β L-Lipschitz on all of $\mathbb{R}$.*

The range of β cannot in general be the range $[0,b)$ of α: $\alpha(x) = 1-\sqrt{1-x}$ on $[0,1)$ approaches 1 more slowly than any Lipschitz function can.

Proof. Extend α to a nondecreasing $g : \mathbb{R} \to [0,b]$ by 0 on the left and b from a on, and set $\beta(x) = a^{-1}\int_0^x g$, so $b' = \beta(a)$. Since $0 \le g \le b$, β is (b/a)-Lipschitz, and with $\beta(0)=0$ this gives $\beta(x) \le (b/a)\,x$. Since $g>0$ on $(0,\infty)$, β is strictly increasing, hence class $\mathcal{K}$ by definition 208. Finally g is nondecreasing, so for $x \in [0,a)$, $\int_0^x g \le x\,\alpha(x) \le a\,\alpha(x)$, i.e. $\beta(x) \le \alpha(x)$. □

Theorem 255 (Subsolutions of the decay system decay)**.** *Let α be class $\mathcal{K}$ on $[0,a)$. Then there is a class $\mathcal{KL}$ function σ with $\sigma(r,0) \le r$ such that: whenever v is continuous on $[t_0,t]$, takes values in $[0,a)$ there, has bounded forward difference quotients, and is a Dini subsolution of the decay system (definition 252),*

$$D^+v(s) \;\le\; -\alpha(v(s)),$$

it obeys $v(t) \le \sigma(v(t_0),\, t-t_0)$.

A differential inequality *with a class $\mathcal{K}$ decay rate therefore yields a class $\mathcal{KL}$ bound — which is what converts a Lyapunov decay estimate into an asymptotic stability statement. No hypothesis is placed on α beyond being class $\mathcal{K}$: the linear-growth condition the construction needs is obtained internally by passing to a Lipschitz minorant.*

Proof. Replace α by a class $\mathcal{K}$ minorant $\beta \le \alpha$ that is Lipschitz near the origin; this only weakens the Dini hypothesis, and supplies the linear bound $\beta(x) \le Lx$ that theorem 253 requires. That theorem gives a class $\mathcal{KL}$ σ whose sections solve $\dot{y} = -\beta(y)$ with $\sigma(r,0)=r$. Since v is a Dini subsolution of the same equation and starts at $\sigma(v(t_0),0)$, theorem 49 bounds v by it — the perturbed solutions that lemma requires coming from theorem 45, whose hypotheses hold because β is globally Lipschitz (lemma 254). □

Chapter 6

Stability of continuous-time systems

Stability notions for an equilibrium of an autonomous continuous-time system $\dot{x} = f(t, x)$ (definition 63); a system with input is first unforced (definition 72). Each notion is defined once, for time-varying systems, with the initial time $t_0 \geq 0$ explicit; a time-invariant system is a special case, and that its notions may be read with $t_0 = 0$ is a theorem rather than a second definition.

Bounds — stability, uniform convergence times, exponential estimates — quantify over trajectories on segments $[t_0, t_1]$, so that a solution escaping in finite time counts against stability instead of being silently excluded. Limits quantify over trajectories on rays $[t_0, \infty)$.

6.1 Definitions

Definition 256 (Stability)**.** The equilibrium x_{eq} of an autonomous continuous-time system $\dot{x} = f(t, x)$ is *stable* when for every $\varepsilon > 0$ and $t_0 \geq 0$ there is $\delta = \delta(\varepsilon, t_0) > 0$ such that every trajectory on a segment $[t_0, t_1]$ with $\|x(t_0) - x_{\text{eq}}\| < \delta$ satisfies $\|x(t) - x_{\text{eq}}\| < \varepsilon$ for all $t \in [t_0, t_1]$.

Quantifying over segments rather than over solutions on $[t_0, \infty)$ means a solution that escapes in finite time counts against stability instead of being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*stable*).

Definition 257 (Instability)**.** The equilibrium is *unstable* when it is not stable (definition 256).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 258 (Uniform stability)**.** The equilibrium is *uniformly stable* when δ in definition 256 can be chosen independently of t_0: $\delta = \delta(\varepsilon)$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly stable*).

Definition 259 (Global uniform stability)**.** The equilibrium is *globally uniformly stable* when it is uniformly stable (definition 258) with a choice of $\delta(\varepsilon)$ satisfying $\delta(\varepsilon) \to \infty$ as $\varepsilon \to \infty$. In particular every trajectory is bounded, so none escapes in finite time.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 260 (Local asymptotic stability)**.** The equilibrium is *locally asymptotically stable* when it is stable (definition 256) and for every $t_0 \geq 0$ there is $c = c(t_0) > 0$ such that the system is forward complete from the ball of radius c about x_{eq} at t_0 (definition 74) and every trajectory on $[t_0, \infty)$ starting in that ball satisfies $x(t) \to x_{\text{eq}}$. Forward completeness keeps solutions that escape in finite time from being silently excluded.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*asymptotically stable*).

Definition 261 (Global asymptotic stability)**.** The equilibrium is *globally asymptotically stable* when it is stable (definition 256), the system is forward complete from every state at every $t_0 \geq 0$ (definition 74), and every trajectory on $[t_0, \infty)$ satisfies $x(t) \to x_{\text{eq}}$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Section 4.1.

Definition 262 (Uniform convergence time)**.** Trajectories starting within c of x_{eq} have a *uniform convergence time*: for every $\eta > 0$ there is $T = T(\eta) > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies $\|x(t) - x_{\text{eq}}\| < \eta$ for $t \in [t_0 + T, t_1]$.

Definition 263 (Uniform asymptotic stability)**.** The equilibrium is *uniformly asymptotically stable* when it is uniformly stable (definition 258) and there is $c > 0$ from which convergence has a uniform convergence time (definition 262).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4 (*uniformly asymptotically stable*).

Definition 264 (Global uniform asymptotic stability)**.** The equilibrium is *globally uniformly asymptotically stable* when it is globally uniformly stable (definition 259) and every $c > 0$ admits a uniform convergence time (definition 262).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.4.

Definition 265 (Local exponential stability)**.** The equilibrium is *locally exponentially stable* when there are $c, k, \gamma > 0$ such that every trajectory on a segment $[t_0, t_1]$, $t_0 \geq 0$, with $\|x(t_0) - x_{\text{eq}}\| < c$ satisfies

$$\|x(t) - x_{\text{eq}}\| \leq k \, \|x(t_0) - x_{\text{eq}}\| \, e^{-\gamma(t - t_0)} \qquad \forall \, t \in [t_0, t_1].$$

The constants do not depend on t_0: exponential stability is uniform, as in the reference.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

Definition 266 (Global exponential stability)**.** The equilibrium is *globally exponentially stable* when the estimate of definition 265 holds for every initial state, with no radius c.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Definition 4.5.

6.2 Implications

Lemma 267 (Instability from a fixed escape radius)**.** *Fix $\varepsilon > 0$. If for every $\delta > 0$ some trajectory on a segment $[0, T]$ starts within δ of x_{eq} and reaches distance at least ε from it, then x_{eq} is unstable (definition 257).*

Proof. Stability at $t_0 = 0$ with tolerance ε gives a δ; the trajectory for that δ contradicts it. □

Lemma 268 (Uniform stability implies stability)**.** *A uniformly stable equilibrium (definition 258) is stable (definition 256).*

Proof. □

Lemma 269 (Global asymptotic stability implies local asymptotic stability)**.** *A globally asymptotically stable equilibrium (definition 261) is locally asymptotically stable (definition 260).*

Proof. □

Lemma 270 (Uniform asymptotic stability implies local asymptotic stability)**.** *A uniformly asymptotically stable equilibrium (definition 263) of a forward-complete system (definition 75) is locally asymptotically stable (definition 260). Forward completeness is needed: bounds on solution segments do not by themselves make the segments extend.*

Proof. Uniform stability gives stability; a uniform convergence time gives convergence of every trajectory on a ray, with the same radius c for every t_0. □

Lemma 271 (Global uniform asymptotic stability implies global asymptotic stability)**.** *A globally uniformly asymptotically stable equilibrium (definition 264) is globally asymptotically stable (definition 261), given forward completeness (definition 75).*

Proof. Every trajectory starts in some ball, and every ball has a uniform convergence time. □

Lemma 272 (Exponential stability implies uniform stability)**.** *A locally exponentially stable equilibrium (definition 265) is uniformly stable (definition 258).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta = \min(c, \varepsilon/k)$ works for every t_0. □

Lemma 273 (Exponential stability implies uniform asymptotic stability)**.** *A locally exponentially stable equilibrium (definition 265) is uniformly asymptotically stable (definition 263).*

Proof. Uniform stability is lemma 272. From the ball of radius c the distance after time T is below $kc\,e^{-\gamma T}$, which is below any $\eta > 0$ for T large, independently of t_0. □

Lemma 274 (Global exponential stability implies local exponential stability)**.** *A globally exponentially stable equilibrium (definition 266) is locally exponentially stable (definition 265).*

Proof. □

Lemma 275 (Global exponential stability implies global uniform asymptotic stability)**.** *A globally exponentially stable equilibrium (definition 266) is globally uniformly asymptotically stable (definition 264).*

Proof. The estimate bounds the distance by k times the initial distance, so $\delta(\varepsilon) = \varepsilon/k$ witnesses global uniform stability. For each radius c the estimate is locally exponential from that ball, which gives a uniform convergence time by lemma 273. □

6.3 Comparison-function characterizations

The uniform notions are equivalent to comparison-function bounds on the trajectories: uniform stability to a class $\mathcal{K}$ bound, uniform asymptotic stability to a class $\mathcal{KL}$ bound. The hard direction is the decay in elapsed time. A uniform convergence time says that for each tolerance η there is a delay after which every trajectory from a ball is inside the η-ball. Read the other way round it bounds how far a trajectory can still be after a given elapsed time, but only through a function of η that may be discontinuous. The results below regularize it into a decay envelope, a function of elapsed time alone that is continuous, positive and vanishing at $+\infty$: the time-decay half of a class $\mathcal{KL}$ bound, whose other half is a uniform class $\mathcal{K}$ bound.

Definition 276 (Uniform class $\mathcal{K}$ bound)**.** The trajectories of s *have the uniform class* $\mathcal{K}$ *bound* $\alpha : [0, a) \to [0, b)$ about x_{eq} when

$$\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in [t_0, t_1]$.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.19).

Definition 277 (Uniform class $\mathcal{K}_\infty$ bound)**.** The trajectories of s *have the uniform class* $\mathcal{K}_\infty$ *bound* α about x_{eq} when $\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Reference: Khalil, *Nonlinear Systems* (3rd ed.), (4.20).

Lemma 278 (Local uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from the ball of radius c (definition 262) and a uniform class $\mathcal{K}$ bound α defined on $[0, a_\alpha)$ (definition 276). Let $0 < a \le c$ with $a < a_\alpha$. Then there is a singular class $\mathcal{L}$ function U (definition 230) such that*

$$\|x(t) - x_{\mathrm{eq}}\| \le U(t - t_0)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in (t_0, t_1]$.

The bound depends on the elapsed time alone: neither on t_0, nor on the trajectory, nor on where in the a-ball it started.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5.*

Proof. Let $\bar{T}(\eta, r)$ be the least delay after which every trajectory starting within r of x_{eq} lies inside the η-ball; it is finite by hypothesis and antitone in η, but possibly discontinuous. Smooth it by the sliding average

$$W(r, \eta) = \frac{2}{\eta} \int_{\eta/2}^{\eta} \left[\bar{T}(s, r) + \frac{r}{\eta} \right] ds,$$

which equals $\frac{2}{\eta} \int_{\eta/2}^{\eta} \bar{T}(s, r)\, ds + r/\eta$. The averaged term is antitone in η and the r/η term is strictly antitone, so $W(r, \cdot)$ is continuous and strictly antitone on $(0, \infty)$; it dominates r/η, hence tends to $+\infty$ as $\eta \to 0^+$, and $\bar{T}(\cdot, r)$ vanishes beyond $\alpha(r)$, hence $W(r, \eta) \to 0$ as $\eta \to \infty$. Inverting it (lemma 224, lemma 225, lemma 226, lemma 227, lemma 228) gives $U = \mathrm{invFunOn}\, W(r, \cdot)\, (0, \infty)$, singular class $\mathcal{L}$. Finally $W(r, U(s)) = s$ and $W \ge \bar{T} + r/\eta$ give $\bar{T}(U(s), r) < s$, so s is itself a valid convergence time to the $U(s)$-ball, which is the claimed bound. □

Lemma 279 (Global uniform decay envelope)**.** *Suppose the trajectories of s have a uniform convergence time from every ball (definition 262) and a uniform class $\mathcal{K}_\infty$ bound (definition 277). Then there is a family $U : (0,\infty) \times (0,\infty) \to \mathbb{R}$ with*

1. *$U(r,s) > 0$ for all $r > 0$ and $s > 0$;*
2. *$U(r,\cdot)$ antitone on $(0,\infty)$ for each $r > 0$;*
3. *$U(r,s) \to 0$ as $s \to \infty$, for each $r > 0$;*
4. *$r \mapsto U(r+1,s)$ monotone on $[0,\infty)$ for each $s > 0$;*
5. *$\|x(t) - x_{\mathrm{eq}}\| \le U(r, t-t_0)$ for every $r > 0$, every trajectory on a segment $[t_0,t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < r$, and every $t \in (t_0,t_1]$.*

Conditions (1)–(4) are exactly the hypotheses of lemma 247; (5) is the decay bound. Unlike the local case (lemma 278) no single envelope can serve every initial state, so the radius remains a parameter.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5 (global case).*

Proof. Every radius is finite, so at radius r the global hypotheses restrict to the local ones (a uniform convergence time from the r-ball, and the class $\mathcal{K}_\infty$ bound read as a class $\mathcal{K}$ bound on $[0, r+1)$). Take $U(r,\cdot)$ to be the envelope of lemma 278 at that radius, namely the inverse of $W(r,\cdot)$. Conditions (1)–(3) and (5) are then that lemma's conclusions, and (4) holds because W is monotone in r — a larger ball has larger optimal convergence times — so inverting reverses nothing. □

Lemma 280 (Uniform convergence from a class $\mathcal{KL}$ bound)**.** *Let β be class $\mathcal{KL}$ on $[0,a) \times [0,\infty)$ (definition 231), and suppose every trajectory on a segment $[t_0,t_1]$, $t_0 \ge 0$, with $\|x(t_0) - x_{\mathrm{eq}}\| < a$ satisfies $\|x(t) - x_{\mathrm{eq}}\| \le \beta(\|x(t_0) - x_{\mathrm{eq}}\|, t - t_0)$ on the segment. Then for every $c < a$, trajectories from the ball of radius c have a uniform convergence time (definition 262).*

Proof. For $c < 0$ no trajectory starts in the ball. Otherwise every deviation is at most $\beta(c, t-t_0)$, by monotonicity of β in its first argument, and $\beta(c,\cdot) \to 0$. □

Lemma 281 (Class $\mathcal{K}$ characterization of stability)**.** *The equilibrium x_{eq} is stable (definition 256) if and only if for each $t_0 \ge 0$ there is a class $\mathcal{K}$ function α_{t_0} on $[0, a_{t_0})$ with $\|x(t) - x_{\mathrm{eq}}\| \le \alpha_{t_0}(\|x(t_0) - x_{\mathrm{eq}}\|)$ for every trajectory on a segment $[t_0,t_1]$ with $\|x(t_0) - x_{\mathrm{eq}}\| < a_{t_0}$, and every $t \in [t_0,t_1]$. A single α for all t_0 is uniform stability (lemma 282).*

Proof. The argument of lemma 282 with the initial time fixed. □

Lemma 282 (Class $\mathcal{K}$ characterization of uniform stability)**.** *The equilibrium x_{eq} is uniformly stable (definition 258) if and only if the trajectories have a uniform class $\mathcal{K}$ bound (definition 276): there is a class $\mathcal{K}$ function α on $[0,a)$ with*

$$\|x(t) - x_{\mathrm{eq}}\| \le \alpha(\|x(t_0) - x_{\mathrm{eq}}\|)$$

for every trajectory on a segment $[t_0,t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\mathrm{eq}}\| < a$, and every $t \in [t_0,t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, first half.*

Proof. Forward: the worst deviation $\omega(r)$ reachable from the r-ball is nondecreasing, vanishes at 0 and is continuous there by uniform stability; a continuous strictly increasing majorant (lemma 250) is the class $\mathcal{K}$ bound. Backward: α is continuous at 0 with $\alpha(0) = 0$, so a small radius δ has $\alpha(\delta) < \varepsilon$. □

Lemma 283 (Class $\mathcal{KL}$ characterization of uniform asymptotic stability)**.** *The equilibrium x_{eq} is uniformly asymptotically stable (definition 263) if and only if there are a class $\mathcal{KL}$ function β on $[0, a) \times [0, \infty)$ (definition 231) with*

$$\|x(t) - x_{\text{eq}}\| \le \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0)$$

for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\text{eq}}\| < a$, and every $t \in [t_0, t_1]$.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, second half.*

Proof. Forward: uniform stability gives a class $\mathcal{K}$ bound α (lemma 282) and, with the uniform convergence time, a decay envelope U (lemma 278); $\beta = \min(\alpha, \sqrt{\alpha\, U})$ is class $\mathcal{KL}$ (theorem 235) and bounds the trajectories. Backward: $\beta(\cdot, 0)$ is continuous at 0, which gives uniform stability, and lemma 280 gives the convergence time. □

Lemma 284 (Class $\mathcal{KL}$ characterization, global case)**.** *The equilibrium x_{eq} is globally uniformly asymptotically stable (definition 264) if and only if there is a $\beta : [0, \infty) \times [0, \infty) \to \mathbb{R}$ that is*

1. *zero at the origin: $\beta(0, s) = 0$ for $s \ge 0$;*
2. *jointly continuous on $[0, \infty) \times [0, \infty)$;*
3. *strictly increasing in r for each fixed $s \ge 0$;*
4. *radially unbounded: $\beta(r, 0) \to \infty$ as $r \to \infty$;*
5. *antitone in s for each fixed $r \ge 0$, with $\beta(r, s) \to 0$ as $s \to \infty$;*

such that $\|x(t) - x_{\text{eq}}\| \le \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0)$ for every trajectory on a segment $[t_0, t_1]$ with $t_0 \ge 0$ and every $t \in [t_0, t_1]$, with no restriction on the initial state.

Condition (4) is what makes the characterization global*: without it β is class $\mathcal{KL}$ on a bounded radius and the statement collapses to the local one (lemma 283).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.5, global case.*

Proof. Forward: global uniform stability gives a class $\mathcal{K}_\infty$ bound, the convergence times from every ball a family of decay envelopes (lemma 279), and lemma 247 combines them. Backward: $\alpha = \beta(\cdot, 0)$ is class $\mathcal{K}_\infty$ and $\delta = \alpha^{-1}$ gives global uniform stability; from the r-ball every deviation is at most $\beta(r, \cdot) \to 0$. □

The same characterizations hold with the bounds on solutions over $[t_0, \infty)$. A bound on every segment restricts to those solutions, so that direction needs nothing more. The converse needs every segment from the ball to extend to such a solution: forward completeness from some ball about x_{eq} locally, and from every state globally.

Lemma 285 (Class $\mathcal{K}$ characterization over $[t_0, \infty)$)**.** *Let the system be forward complete from some ball $B(x_{\text{eq}}, c)$ at every $t_0 \ge 0$ (definition 74). Then x_{eq} is uniformly stable if and only if there is a class $\mathcal{K}$ function α on $[0, a)$ with $\|x(t) - x_{\text{eq}}\| \le \alpha(\|x(t_0) - x_{\text{eq}}\|)$ for every solution x on $[t_0, \infty)$ with $t_0 \ge 0$ and $\|x(t_0) - x_{\text{eq}}\| < a$, and every $t \ge t_0$. The forward direction holds without completeness.*

Proof. lemma 282, with the bound moved between segments and solutions on $[t_0, \infty)$. Forward: restrict a solution to $[t_0, t]$. Converse: shrink the radius to $a' = \min(c, a/2)$, restricting α (definition 209); a segment starting within a' extends by completeness to a solution on $[t_0, \infty)$, which the bound covers. □

Lemma 286 (Class $\mathcal{KL}$ characterization over $[t_0, \infty)$)**.** *Let the system be forward complete from some ball $B(x_{\text{eq}}, c)$ at every $t_0 \geq 0$ (definition 74). Then x_{eq} is uniformly asymptotically stable if and only if there is a class $\mathcal{KL}$ function β on $[0, a) \times [0, \infty)$ with $\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0)$ for every solution x on $[t_0, \infty)$ with $t_0 \geq 0$ and $\|x(t_0) - x_{\text{eq}}\| < a$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 283, with the bound moved between segments and solutions on $[t_0, \infty)$ as in lemma 285, shrinking the radius to $\min(c, a)$ (definition 239). □

Lemma 287 (Class $\mathcal{KL}$ characterization over $[t_0, \infty)$, global case)**.** *Let the system be forward complete (definition 75). Then x_{eq} is globally uniformly asymptotically stable if and only if there is a β with the properties of lemma 284 and $\|x(t) - x_{\text{eq}}\| \leq \beta(\|x(t_0) - x_{\text{eq}}\|, t - t_0)$ for every solution x on $[t_0, \infty)$ with $t_0 \geq 0$, and every $t \geq t_0$. The forward direction holds without completeness.*

Proof. lemma 284, with the bound moved between segments and solutions on $[t_0, \infty)$ as in lemma 285. □

6.4 Lyapunov functions

A Lyapunov function is a certificate: its existence implies a stability notion. The certificates are time-varying, $V(t, x)$, and named by what they certify. What separates them is which bounds hold, not whether time appears: a positive definite (definition 25) lower bound gives stability, and a positive definite upper bound as well (decrescence) is what makes the conclusion uniform. A time-invariant $V(x)$ is its own lower and upper bound, so it is a uniform certificate as soon as it decreases along the system.

Definition 288 (Lyapunov function)**.** A differentiable $V : \mathbb{R} \times E \to \mathbb{R}$ is a *Lyapunov function* on a neighbourhood D of x_{eq} when, for all $t \geq 0$ and $x \in D$,

$$V(t, x_{\text{eq}}) = 0, \qquad W_1(x) \leq V(t, x), \qquad \dot{V}(t, x) \leq 0,$$

for some W_1 positive definite on D (definition 25), with $\dot{V}$ the derivative along the system (definition 86). It certifies stability (definition 256).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 289 (Uniform Lyapunov function)**.** A Lyapunov function (definition 288) that is also *decrescent*: $V(t, x) \leq W_2(x)$ for all $t \geq 0$, $x \in D$, for some W_2 positive definite on D. The upper bound is what makes the conclusion uniform: it certifies uniform stability (definition 258).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.8.

Definition 290 (Strict uniform Lyapunov function)**.** A uniform Lyapunov function (definition 289) with $\dot{V}(t, x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. It certifies uniform asymptotic stability (definition 263).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.9.

Definition 291 (Exponential Lyapunov function)**.** A differentiable $V : \mathbb{R}\times E \to \mathbb{R}$ is an *exponential Lyapunov function* on a neighbourhood D of x_{eq} when there are $k_1, k_2, k_3, a > 0$ with, for all $t \geq 0$ and $x \in D$,

$$k_1\|x - x_{\text{eq}}\|^a \leq V(t,x) \leq k_2\|x - x_{\text{eq}}\|^a, \qquad \dot{V}(t,x) \leq -k_3\|x - x_{\text{eq}}\|^a.$$

It certifies local exponential stability (definition 265).

Reference: Khalil, *Nonlinear Systems* (3rd ed.), Theorem 4.10.

Lemma 292 (Time-invariant Lyapunov functions are uniform)**.** *Let $V : E \to \mathbb{R}$ be differentiable and positive definite on a neighbourhood D of x_{eq} (definition 25), with $\nabla V(x) \cdot f(t,x) \leq 0$ for all $t \geq 0$, $x \in D$. Then $(t,x) \mapsto V(x)$ is a uniform Lyapunov function (definition 289).*

Proof. Take $W_1 = W_2 = V$. The derivative along the system of a function of x alone is its directional derivative along the vector field. □

Lemma 293 (Strict time-invariant Lyapunov functions are strict uniform)**.** *Let $V : E \to \mathbb{R}$ be differentiable and positive definite on a neighbourhood D of x_{eq}, with $\nabla V(x) \cdot f(t,x) \leq -W_3(x)$ for all $t \geq 0$, $x \in D$, for some W_3 positive definite on D. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 290).*

Proof. As for lemma 292; $-W_3 \leq 0$ gives the non-strict decrease. □

Lemma 294 (Strict decrease suffices for a time-invariant system)**.** *Let the state equation be $\dot{x} = f(x)$ with f continuous and $f(x_{\text{eq}}) = 0$, and let $V : E \to \mathbb{R}$ be C^1 and positive definite on a neighbourhood D of x_{eq} with $\nabla V(x) \cdot f(x) < 0$ for $x \in D \setminus \{x_{\text{eq}}\}$. Then $(t,x) \mapsto V(x)$ is a strict uniform Lyapunov function (definition 290).*

Proof. Take $W_3 = -\nabla V \cdot f$: continuous since V is C^1 and f is continuous, zero at x_{eq} since $f(x_{\text{eq}}) = 0$, and positive elsewhere on D; apply lemma 293. □

6.5 Lyapunov's direct method

For a time-varying V the derivative along the system acquires an explicit time term, and positive definiteness of $V(t,\cdot)$ on its own buys nothing uniform in t_0: V could flatten as $t \to \infty$. Sandwiching V between two positive-definite functions of the state alone is what restores uniformity. Only trajectories on segments $[t_0, t_1]$ are used, so no solution is assumed to exist for all time.

Theorem 295 (Lyapunov's uniform stability theorem)**.** *If V is a uniform Lyapunov function for x_{eq} (definition 289) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly stable (definition 258).*

The sandwich $W_1(x) \leq V(t,x) \leq W_2(x)$ is what makes the conclusion uniform in t_0: W_1 and W_2 do not depend on t, so neither does the $\delta(\varepsilon)$ extracted from them.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.8.*

Proof. Given ε, work on a radius $r \leq \varepsilon$ whose closed ball lies in the region of the certificate. Bound W_1 below and W_2 above by class $\mathcal{K}$ functions α_1, α_2 on it (theorem 251), fix a level d below both ranges, and take $\delta = \alpha_2^{-1}(d)$: starting within δ of x_{eq} forces $V(t_0, x(t_0)) < d$. To leave the r-ball the trajectory would first cross a sphere on which $W_1 > d$; up to that time V has not increased (lemma 54), so it is still below d, a contradiction. □

Theorem 296 (Lyapunov's uniform asymptotic stability theorem)**.** *If V is a strict uniform Lyapunov function for x_{eq} (definition 290) on a system with state space $\mathbb{R}^n$, then x_{eq} is uniformly asymptotically stable (definition 263).*

A strictly negative $\dot{V}$ alone would not do: W_3 must be bounded away from zero on each annulus, which is exactly what positive definiteness of a function of x alone buys.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.9. Khalil's two "moreover" clauses are not part of this conclusion: the explicit class $\mathcal{KL}$ bound, which the proof constructs, and the global case.*

Proof. Uniform stability is theorem 295. For convergence, take class $\mathcal{K}$ bounds $\alpha_1, \alpha_2, \alpha_3$ for W_1, W_2, W_3 on a ball in the region of the certificate, and the trapping radius c of that proof. Along a trajectory from the c-ball, $\dot{V} \le -W_3(x) \le -\alpha_3(\|x - x_{\text{eq}}\|) \le -\alpha_3(\alpha_2^{-1}(V))$, so $v(t) = V(t, x(t))$ satisfies $D^+v \le -\alpha(v)$ with $\alpha = \alpha_3 \circ \alpha_2^{-1}$ of class $\mathcal{K}$. theorem 255 gives a class $\mathcal{KL}$ σ with $v(t) \le \sigma(v(t_0), t - t_0)$; unwinding the sandwich,

$$\|x(t) - x_{\text{eq}}\| \le \alpha_1^{-1}(\sigma(\alpha_2(\|x(t_0) - x_{\text{eq}}\|), t - t_0)),$$

a class $\mathcal{KL}$ bound. Conclude by lemma 280. □

In the global theorem compact sublevel sets of V are the hypothesis; for a continuous V they follow from radial unboundedness.

Lemma 297 (Sublevel sets of a radially unbounded function are compact)**.** *Let E be a proper normed space. If $V : E \to \mathbb{R}$ is continuous and radially unbounded ($V(x) \to \infty$ as $\|x\| \to \infty$), then every sublevel set $\{x \mid V(x) \le c\}$ is compact.*

Proof. Closed: it is $V^{-1}((-\infty, c])$, with V continuous. Bounded: radial unboundedness gives a compact K off which $V > c$, so the sublevel set lies in K. Compact: closed and bounded in a proper space. □

Theorem 298 (Lyapunov's global asymptotic stability theorem)**.** *Let the state equation be $\dot{x} = f(x)$ with f continuous, on $\mathbb{R}^n$. Let $V : \mathbb{R}^n \to \mathbb{R}$ be C^1, positive definite about x_{eq} on all of $\mathbb{R}^n$ (definition 25), with compact sublevel sets $\{V \le c\}$ (equivalently, radially unbounded) and $\nabla V(x) \cdot f(x) < 0$ for $x \ne x_{\text{eq}}$. Then x_{eq} is globally uniformly asymptotically stable (definition 264).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.2.*

Proof. V does not increase along trajectories: $\nabla V \cdot f \le 0$ everywhere, since $\nabla V(x_{\text{eq}}) = 0$ at the minimum. So a trajectory starting below every value V takes outside the ε-ball never leaves it. Take $\delta(\varepsilon)$ the supremum of the radii $\rho \le \varepsilon$ whose closed ball lies strictly below those values: it is positive, since V is bounded below by a positive constant outside the ε-ball and small near x_{eq}, and it tends to ∞, since V is bounded on each ball and large far out. This is global uniform stability.

For convergence from the c-ball, trajectories stay in the compact set $\{V \le M\}$, $M = \max_{\overline{B}(x_{\text{eq}}, c)} V$, on which, outside the $\delta(\eta)$-ball, $\nabla V \cdot f \le -\gamma < 0$. So V falls at rate γ while the trajectory is outside, and since $V \ge 0$ it enters the $\delta(\eta)$-ball by time $t_0 + M/\gamma + 1$; from then on it stays in the η-ball. □

Theorem 299 (Lyapunov's exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} (definition 291), then x_{eq} is locally exponentially stable (definition 265): near x_{eq},*

$$\|x(t) - x_{\text{eq}}\| \le \left(\frac{k_2}{k_1}\right)^{1/a} \|x(t_0) - x_{\text{eq}}\| \, e^{-(k_3/(ak_2))(t - t_0)}.$$

The state space is any normed space.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10.*

Proof. The bounds give $\dot V \le -(k_3/k_2)V$, so $e^{(k_3/k_2)t}V(t,x(t))$ does not increase and $V(t,x(t)) \le V(t_0,x(t_0))\,e^{-(k_3/k_2)(t-t_0)}$ while the trajectory stays in D; the sandwich and an a-th root give the estimate. To stay in D, start within $r/(K+1)$ of x_{eq}, where $\overline{B}(x_{\mathrm{eq}},r) \subseteq D$ and $K = (k_2/k_1)^{1/a}$: at the first time the trajectory reached the sphere of radius r the estimate would put it strictly inside. □

Theorem 300 (Lyapunov's global exponential stability theorem)**.** *If V is an exponential Lyapunov function for x_{eq} on the whole space (definition 291 with $D = E$), then x_{eq} is globally exponentially stable (definition 266).*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.10, global case.*

Proof. The estimate of theorem 299 holds on every trajectory segment, since every segment stays in $D = E$. □

6.6 LaSalle's invariance principle

When $\nabla V \cdot f \le 0$ only, V may stall, and Lyapunov's theorems give stability but not convergence. LaSalle's principle recovers convergence for a time-invariant system: a trajectory confined to a compact set converges to its ω-limit set, on which V is constant, and forward invariance of that set turns constancy of V into $\nabla V \cdot f = 0$.

Lemma 301 (The ω-limit set is forward invariant)**.** *Let $\dot x = f(x)$ with f Lipschitz, and let φ be a trajectory on $[t_0,\infty)$. Then its ω-limit set $\omega(\varphi)$ (definition 57) is forward invariant (definition 85).*

Neither boundedness of φ nor compactness of an enclosing set is needed; those hypotheses of Khalil's Lemma 4.1 are what make $\omega(\varphi)$ nonempty and compact, not what make it invariant.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Lemma 4.1.*

Proof. Let ψ be a trajectory on $[a,b]$ with $p = \psi(a) \in \omega(\varphi)$, and fix $t \in [a,b]$. Given $\varepsilon > 0$, put $C = e^{K(b-a)}$ and pick, by the definition of the ω-limit set, arbitrarily large r with $\|\varphi(r) - p\| < \varepsilon/C$. The time-shifted curve $u(\cdot) = \varphi(\cdot + r - a)$ solves the same time-invariant equation on $[a,b]$ with $u(a) = \varphi(r)$, so Grönwall gives

$$\|\varphi(r+t-a) - \psi(t)\| = \|u(t) - \psi(t)\| \le \|\varphi(r) - p\|\, e^{K(t-a)} < \varepsilon.$$

Since r may be taken arbitrarily large, so may $r + t - a$; hence φ is frequently within ε of $\psi(t)$, i.e. $\psi(t) \in \omega(\varphi)$.

Khalil argues instead with the flow map and the semigroup identity, which presumes a globally defined flow; comparing the two solutions directly needs only uniqueness, in its quantitative Grönwall form. □

Lemma 302 ($\dot V$ vanishes on the ω-limit set)**.** *Let $\dot x = f(x)$ with f Lipschitz, Ω compact and forward invariant, $V \in C^1$ with $\nabla V \cdot f \le 0$ on Ω, and φ a trajectory on $[t_0,\infty)$ with $\varphi(t_0) \in \Omega$. Then $\nabla V(y) \cdot f(y) = 0$ for every $y \in \omega(\varphi)$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), proof of Theorem 4.4.*

Proof. $V(\varphi(t))$ is nonincreasing and bounded below on the compact Ω, so it converges to some L, and $V \equiv L$ on $\omega(\varphi)$ (lemma 60). Through $y \in \omega(\varphi)$ take a solution ψ on $[0,1]$ (Picard–Lindelöf, theorem 44); it stays in $\omega(\varphi)$ by lemma 301, so $V \circ \psi \equiv L$ and its derivative at 0, which is $\nabla V(y) \cdot f(y)$, vanishes. □

Theorem 303 (LaSalle's invariance principle)**.** *Let $\dot{x} = f(x)$ with f Lipschitz, Ω compact and forward invariant (definition 85), and $V \in C^1$ with $\nabla V \cdot f \leq 0$ on Ω. Put*

$$E = \{x \in \Omega : \nabla V(x) \cdot f(x) = 0\},$$

and let M contain every forward invariant subset of E. Then every trajectory φ on $[t_0, \infty)$ with $\varphi(t_0) \in \Omega$ satisfies $\varphi(t) \to M$ as $t \to \infty$.

Khalil takes M to be the largest invariant *subset of E. Here M must absorb every* forward *invariant subset, a larger family, so the hypothesis on M is correspondingly stronger: backward invariance of $\omega(\varphi)$ would need solutions through its points to extend backwards in time.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.4.*

Proof. The ω-limit set lies in Ω, which is closed and contains the trajectory (lemma 61), and $\nabla V \cdot f$ vanishes on it (lemma 302), so $\omega(\varphi) \subseteq E$. It is forward invariant (lemma 301), so $\omega(\varphi) \subseteq M$. A trajectory confined to a compact set converges to every set containing its ω-limit set (lemma 62). □

Theorem 304 (Barbashin's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on a neighbourhood D of x_{eq} (definition 25) with $\nabla V \cdot f \leq 0$ on D. Suppose that for some $c > 0$ the sublevel set $\Omega_c = \{V \leq c\}$ is contained in D, compact and forward invariant, and that $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x \in \Omega_c : \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is locally asymptotically stable (definition 260).*

The hypothesis on E is the one a user of LaSalle discharges: solve $\nabla V \cdot f = 0$ and check that no solution can stay there but the equilibrium.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.1.*

Proof. Stability: V is a uniform Lyapunov function (lemma 292, theorem 295). Forward completeness: f is Lipschitz (theorem 78). Convergence: near x_{eq}, $V < c$, so a trajectory starting there starts in Ω_c, and theorem 303 with $M = \{x_{\text{eq}}\}$ applies. □

Theorem 305 (Krasovskii's theorem)**.** *Let $\dot{x} = f(x)$ on $\mathbb{R}^n$ with f Lipschitz, and $V \in C^1$ positive definite on $\mathbb{R}^n$, with compact sublevel sets (equivalently, radially unbounded) and $\nabla V \cdot f \leq 0$ everywhere. Suppose that for every c, $\{x_{\text{eq}}\}$ contains every forward invariant subset of $\{x : V(x) \leq c,\ \nabla V(x) \cdot f(x) = 0\}$. Then x_{eq} is globally asymptotically stable (definition 261).*

Unlike Barbashin's theorem (theorem 304) no forward invariant set is assumed: every sublevel set is compact, and $\nabla V \cdot f \leq 0$ keeps a trajectory in the one through its initial state.

Reference: Khalil, Nonlinear Systems *(3rd ed.), Corollary 4.2.*

Proof. Stability and forward completeness as in Barbashin's theorem. The sublevel set $\{V \leq V(x(t_0))\}$ is compact and, since V does not increase along trajectories, forward invariant; apply theorem 303 with $M = \{x_{\text{eq}}\}$. □

6.7 Instability

Instability is the failure of stability, and the simplest witness is a trajectory that leaves a fixed neighbourhood from arbitrarily close initial states (lemma 267). Chetaev's theorem produces such trajectories from a function that grows along the system near the equilibrium.

Theorem 306 (Exponential Chetaev instability criterion)**.** *Suppose the vector field f and certificate V are C^1, V is quadratically bounded on a ball, its positive set accumulates at the base point, and its Lie derivative satisfies $\dot{V} \geq 2\alpha V$ there for some $\alpha > 0$. Then the base point is unstable (definition 257) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Hahn, Stability of Motion *(Chetaev's instability method).*

Proof. Globalize the vector field by a smooth cutoff, integrate the differential inequality on a sufficiently long finite segment, and stop the curve at its first crossing of the certificate ball. □

Chapter 7

Stability of discrete-time systems

The notions of the previous chapter for an autonomous discrete-time system $x^+ = f(k, x)$ (definition 92), under the same names. Discrete time is simpler in one respect: trajectories always exist and are unique (**??**, **??**), so every notion quantifies over trajectories on $[k_0, \infty)$ and no forward completeness is needed anywhere. Exponential estimates are geometric, ρ^{k-k_0} with $0 \leq \rho < 1$, in place of $e^{-\gamma(t-t_0)}$.

7.1 Definitions

Ultimate boundedness is a property of the system near a point that need not be an equilibrium: trajectories end up, and stay, within a prescribed distance of it.

7.2 Implications

Chapter 8

Stability of linear systems

The stability notions of a linear system are those of definition 256 and the definitions after it, applied to the unforced system (definition 72): there is no separate notion of stability for linear systems. What linearity adds is that each notion becomes a condition on the state transition matrix, and for constant A a condition on its eigenvalues.

8.1 Linear time-varying systems

Hespanha states Lyapunov stability for the continuous-time linear time-varying system $\dot{x} = A(t)x$ as four conditions on its homogeneous state response $x(t) = \Phi(t, t_0)x_0$. None of them is taken as a definition here: each is proved *equivalent* to the corresponding stability predicate of the unforced system, so the linear track and the nonlinear track share one notion of stability rather than two.

Two things stand between the textbook conditions and the predicates, and both are settled in this section. The predicates quantify over trajectories where the textbook quantifies over initial conditions, which the trajectory characterization below closes in both directions; and the predicates bound the response from a *ball* where the textbook bounds it from every initial state, which homogeneity closes. The second is why a linear system has no nontrivial region of attraction, recorded as lemma 313.

8.1.1 Hespanha's Definition 8.1

Lemma 307 (Stability as a bound on the state transition matrix)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\,x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is a stable equilibrium (definition 256) of this system if and only if for every $t_0 \geq 0$ there is C with

$$\|\Phi(t, t_0)\,x\| \leq C\,\|x\|, \qquad \forall\, t \geq t_0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. ($\Rightarrow$) Apply stability at $\varepsilon = 1$ to get $\delta(1, t_0)$. Every response from the ball of radius δ stays inside the unit ball, and scaling turns that into the bound with $C = 2/\delta$.

($\Leftarrow$) Given ε, take $\delta = \varepsilon/(C+1)$. A trajectory is its own response (theorem 173), so $\|\varphi(t)\| \leq C\|\varphi(t_0)\| < \varepsilon$. □

Theorem 308 (Lyapunov stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is a stable equilibrium (definition 256) of this system if and only if, for every $t_0 \geq 0$ and every initial condition x_0, the homogeneous state response $x(t) = \Phi(t, t_0)\, x_0$ is bounded on $[t_0, \infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1).*

Proof. The state-transition form is lemma 307. A bound on Φ gives the response bound $C\|x_0\|$; conversely a family of operators bounded at every point of a Banach space is norm-bounded, by the uniform boundedness principle. □

Theorem 309 (Uniform stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is a uniformly stable equilibrium (definition 258) of this system if and only if there is a single C with

$$\|\Phi(t, t_0)\, x\| \leq C\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(1), with the margin independent of t_0.*

Proof. Identical to lemma 307, with the margin δ and the bound C both carried outside the quantifier on t_0. □

Theorem 310 (Asymptotic stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is an asymptotically stable equilibrium (definition 260) of this system if and only if, for every $t_0 \geq 0$ and every x_0, the homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is bounded on $[t_0, \infty)$ and tends to 0 as $t \to \infty$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(2).*

Proof. The stability conjunct is theorem 308. For attractivity: an arbitrary x_0 is rescaled into the attractivity ball, and the limit scales back, since the response is linear in the initial state; conversely every trajectory is a response (theorem 173), so convergence of the responses is convergence of the trajectories, from any radius. □

Theorem 311 (Global exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is a globally exponentially stable equilibrium (definition 266) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. Both directions are the trajectory characterization plus homogeneity: the exponential estimate along every trajectory is the estimate applied to the initial state, and conversely the estimate on any ball scales to the estimate everywhere. □

Theorem 312 (Exponential stability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is an exponentially stable equilibrium (definition 265) of this system if and only if there are $k, \gamma > 0$ with

$$\|\Phi(t, t_0)\, x\| \leq k\, e^{-\gamma(t-t_0)}\, \|x\|, \qquad \forall\, t \geq t_0 \geq 0,\ \forall\, x.$$

The condition is the one in theorem 311: the radius in the local definition carries no information here.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(3).*

Proof. As in the global case; the radius c of the local definition is absorbed by the scaling argument, and $c = 1$ serves in the converse. □

Lemma 313 (Local exponential stability is global, for a linear system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is an exponentially stable equilibrium (definition 265) of this system if and only if it is globally exponentially stable (definition 266).

Proof. Both are equivalent to the same estimate, theorem 312 and theorem 311. □

Theorem 314 (Instability of a linear time-varying system)**.** *Consider the linear time-varying system*

$$\dot{x} = A(t)\, x,$$

with A continuous, and write $\Phi(t, t_0)$ for its state transition matrix (definition 141).

The origin is unstable (definition 257) for this system if and only if there are $t_0 \geq 0$ and x_0 whose homogeneous response $t \mapsto \Phi(t, t_0)x_0$ is unbounded on $[t_0, \infty)$.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Definition 8.1(4).*

Proof. The negation of theorem 308. □

8.2 Linear time-invariant systems

For constant A the state transition matrix is $e^{A(t-t_0)}$ (theorem 179), and the stability conditions of the previous section collapse to a condition on the eigenvalues of A alone. This is Hespanha's Theorem 8.1.

Clauses (1) and (4) of that theorem — marginal stability and instability — ask that the Jordan blocks at imaginary-axis eigenvalues be 1×1. They are not recorded here: they need a Jordan normal form, which is a project rather than a lemma.

8.2.1 The Hurwitz condition and its rate-indexed refinement

For a real square matrix A, Hurwitz stability is expressed through the complex eigenpairs of its entrywise complexification, over an abstract index type. The rate-indexed predicate records a strict spectral margin. Ordinary Hurwitz stability is the zero-rate case, but is stated separately rather than derived, so that it unfolds directly to $\mathrm{Re}(\mu) < 0$.

Definition 315 (Hurwitz matrix)**.** A real square matrix A is *Hurwitz* when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < 0$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Definition 316 (Hurwitz with a prescribed decay rate)**.** A real square matrix A is *Hurwitz with decay rate* α when every eigenpair (μ, v) of its complexification, with $v \neq 0$, satisfies $\mathrm{Re}(\mu) < -\alpha$.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1.

Theorem 317 (Rate-Hurwitz via a spectral shift)**.** *A real matrix A is Hurwitz with decay rate α if and only if the spectrally shifted matrix $A + \alpha I$ is Hurwitz.*

Reference: João P. Hespanha, Linear Systems Theory *(2nd ed.), continuous-time stability criterion. This spectral-shift corollary is proved directly from eigenpairs.*

Proof. A complex eigenvalue μ of A becomes $\mu + \alpha$ after the shift, and $\mathrm{Re}(\mu) < -\alpha$ is equivalent to $\mathrm{Re}(\mu + \alpha) < 0$. □

Theorem 318 (A Hurwitz matrix has a positive spectral margin)**.** *If A is Hurwitz then there is $c > 0$ such that every eigenvalue μ of A satisfies $\mathrm{Re}(\mu) < -c$.*

Proof. The spectrum is compact and nonempty, so Re attains a maximum on it at some μ_0, which is an eigenvalue and hence has $\mathrm{Re}(\mu_0) < 0$. Take $c = -\mathrm{Re}(\mu_0)/2$. □

The exponential decay estimates rest on a general Banach-algebra fact, stated with no matrices in it: an element of spectral radius below one has a contractive power.

Lemma 319 (A contractive power from a spectral radius below one)**.** *Let $\mathbb{A}$ be a nontrivial complete normed algebra over $\mathbb{C}$ and let $a \in \mathbb{A}$ have spectral radius $r(a) < 1$. Then there is an $m > 0$ with $\|a^m\| < 1$.*

Reference: Rudin, Functional Analysis *(Gelfand's spectral-radius formula).*

Proof. Gelfand's formula gives $\|a^m\|^{1/m} \to r(a)$, so $\|a^m\|^{1/m} < 1$ for all large m; pick any such m with $m \neq 0$. Were $\|a^m\| \geq 1$, raising to the positive power $1/m$ would give $\|a^m\|^{1/m} \geq 1$, a contradiction. □

The two predicates are related by a spectral shift (theorem 317), and they agree at rate zero.

Lemma 320 (Hurwitz is rate zero)**.** *A real square matrix A is Hurwitz (definition 315) if and only if it is Hurwitz with decay rate 0 (definition 316).*

Proof. Both unfold to a condition on $\operatorname{Re}\mu$, the first to $\operatorname{Re}\mu < 0$ and the second to $\operatorname{Re}\mu < -0$. □

Lemma 321 (A certified decay rate may be weakened)**.** *If A is Hurwitz with decay rate α (definition 316) and $\beta \le \alpha$, then A is Hurwitz with decay rate β.*

Proof. Each eigenvalue satisfies $\operatorname{Re}\mu < -\alpha \le -\beta$. □

Lemma 322 (Zero-dimensional matrices are vacuously Hurwitz)**.** *If the index type is empty then every A is Hurwitz with every decay rate α (definition 316).*

This records the convention chosen for the degenerate case: the predicate quantifies over nonzero eigenvectors, and in dimension zero there are none.

Proof. There is no nonzero vector, so the hypothesis of the predicate is never met. □

Lemma 323 (The scalar example)**.** *For $\alpha < \gamma$, the 1×1 matrix $(-\gamma)$ is Hurwitz with decay rate α (definition 316).*

Included as a concrete check on the sign convention of the definition.

Proof. The only eigenvalue is $-\gamma$, and $\operatorname{Re}(-\gamma) = -\gamma < -\alpha$. □

The Hurwitz condition bounds each eigenvalue away from the imaginary axis one at a time; because the spectrum is compact, that upgrades to a uniform margin (theorem 318). This is what turns a qualitative hypothesis into a quantitative rate, and it is used twice below.

8.2.2 Equivalent analytic conditions

Nothing in this subsection mentions a system: the Hurwitz condition, decay of e^{At}, and an explicit exponential envelope for $\|e^{At}\|$ are shown to be the same condition on the matrix. Two generic facts about exponential envelopes are recorded first.

Lemma 324 (The decay envelope vanishes)**.** *For $\gamma > 0$ and any constants K, t_0,*

$$\lim_{t\to\infty} K\, e^{-\gamma(t-t_0)} = 0.$$

Proof. The affine map $t \mapsto -\gamma(t - t_0)$ tends to $-\infty$ because $-\gamma < 0$; compose with exp and scale by K. □

Lemma 325 (Squeeze against an exponentially decaying envelope)**.** *Let E be a normed space, $f : \mathbb{R} \to E$, and $\gamma > 0$. If eventually*

$$\|f(t)\| \le k\, e^{-\gamma(t-t_0)}\, c,$$

then $f(t) \to 0$ as $t \to \infty$.

Stated over an arbitrary normed space because it is applied to both matrix- and vector-valued quantities.

Proof. The squeeze theorem: $\|f\|$ is bounded below by 0 and above by an envelope that tends to 0 (lemma 324, after reassociating $k \cdot E \cdot c = (kc) \cdot E$). □

Theorem 326 (A Hurwitz matrix has an exponentially decaying exponential). *If A is Hurwitz (definition 315) then there are $k > 0$ and $\gamma > 0$ with*

$$\|e^{At}\| \le k\, e^{-\gamma t}, \qquad \forall\, t \ge 0.$$

Proof. A contracting block. The spectral radius of $e^{A_{\mathbb{C}}}$ is below 1 (lemma 193), so by Gelfand's formula (lemma 319) some integer time $p > 0$ has $\|e^{Ap}\| < 1$. Fix c with $\|e^{Ap}\| \le c < 1$, kept away from 0 so that $\log c$ exists.

One block. $\|e^{Ar}\|$ is continuous, hence bounded by some $M \ge 1$ on the compact interval $[0, p]$.

Spreading. Write $t = qp + r$ with $q \in \mathbb{N}$ and $r \in [0, p)$. The exponents commute, so $e^{At} = e^{Ar}(e^{Ap})^q$, and submultiplicativity gives $\|e^{At}\| \le Mc^q$ by induction on q. Setting $\gamma = -\log c/p$ turns c^q into $e^{-\gamma qp}$, and $qp \ge t - p$ costs one further factor c^{-1}. The bound is $\|e^{At}\| \le (M/c)\, e^{-\gamma t}$.

The contraction is taken in the L^∞ operator norm, which is what makes it usable against $\|Ax\|_\infty \le \|A\|_\infty \|x\|_\infty$ in theorem 327.

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.1. □

Theorem 327 (Exponential decay of the state under a Hurwitz matrix). *If A is Hurwitz (definition 315) then there are $k > 0$ and $\gamma > 0$ with*

$$\|e^{At}x\| \le k\, e^{-\gamma t}\, \|x\|, \qquad \forall\, t \ge 0,\ \forall\, x.$$

This is the form the Lyapunov stability predicates consume.

Proof. theorem 326 bounds the matrix; one application of $\|Ax\|_\infty \le \|A\|_\infty \|x\|_\infty$ transfers it to the state. This is the step at which the choice of the L^∞ operator norm earns its keep. □

Theorem 328 (Decay of the matrix exponential forces the Hurwitz condition). *If $e^{At} \to 0$ as $t \to \infty$ then A is Hurwitz (definition 315).*

Stated from mere convergence rather than from an exponential bound, so that it serves clause (2) and clause (3) of Theorem 8.1 alike.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1.*

Proof. Suppose $\operatorname{Re}\mu \ge 0$ for some eigenpair (μ, v) with $v \ne 0$. Complexification is continuous, so $e_{\mathbb{C}}^{At}v \to 0$ as well. But v is an eigenvector of $tA_{\mathbb{C}}$ with eigenvalue $t\mu$, hence of its exponential (lemma 188), so

$$\|e_{\mathbb{C}}^{At}v\| = |e^{t\mu}|\, \|v\| = e^{t \operatorname{Re}\mu}\|v\| \ge \|v\| > 0$$

for every $t \ge 0$, contradicting convergence to 0. □

Theorem 329 (Pointwise decay of the state forces the Hurwitz condition). *If $e^{At}x \to 0$ as $t \to \infty$ for every real state x, then A is Hurwitz (definition 315).*

This is the composite that both clauses of Theorem 8.1 reach for: the stability predicates speak about states, not about the matrix.

Proof. Feeding the hypothesis the standard basis vectors returns the columns of e^{At}, and convergence in the matrix space is entrywise, so $e^{At} \to 0$; apply theorem 328. □

Theorem 330 (Hurwitz is decay of the matrix exponential)**.** *A real square matrix A is Hurwitz (definition 315) if and only if $e^{At} \to 0$ as $t \to \infty$.*

Proof. Forwards, theorem 326 supplies an exponentially decaying envelope and lemma 325 squeezes; backwards is theorem 328. □

Theorem 331 (Hurwitz is an exponential envelope for the matrix exponential)**.** *A real square matrix A is Hurwitz (definition 315) if and only if there are $k > 0$ and $\gamma > 0$ with $\|e^{At}\| \le k\, e^{-\gamma t}$ for all $t \ge 0$.*

Recorded separately from theorem 330 because the two are what clauses (2) and (3) of Theorem 8.1 respectively need: bare convergence for asymptotic stability, an explicit rate for exponential stability.

Proof. Forwards is theorem 326. Backwards, the envelope forces $e^{At} \to 0$ (lemma 325), so theorem 330 applies. □

8.2.3 Hespanha's Theorem 8.1

Theorem 332 (Asymptotic stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an asymptotically stable equilibrium (definition 260) if and only if A is Hurwitz (definition 315), that is, every eigenvalue of A has strictly negative real part.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(2).*

Proof. For constant A the state transition matrix is $e^{A(t-t_0)}$ (theorem 179), so theorem 310 turns the statement into one about $e^{At}x_0$.

If the response converges then A is Hurwitz (theorem 329). Conversely a Hurwitz A gives an exponentially decaying envelope for $\|e^{At}x_0\|$ (theorem 327); the envelope is bounded by $k\|x_0\|$, which supplies the stability half, and tends to zero (lemma 324), which supplies the attractivity half. □

Theorem 333 (Exponential stability of a linear time-invariant system)**.** *Consider the time-invariant system $\dot{x} = Ax$ with A a real square matrix.*

The origin is an exponentially stable equilibrium (definition 265) if and only if A is Hurwitz (definition 315).

Together with theorem 332 this says the two notions coincide for a time-invariant system — a genuine feature of constant A, not of linearity: for time-varying $A(t)$ they differ, as $\dot{x} = -x/(1+t)$ shows.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.1(3).*

Proof. As for theorem 332, but against theorem 312, which already asks for an exponential envelope. The forward direction squeezes $\|e^{At}x\|$ to zero under that envelope and applies theorem 329; the converse is theorem 327 re-anchored from 0 to t_0. □

8.2.4 The Lyapunov equation

Definition 334 (Continuous Lyapunov equation)**.** For real square matrices A, P and Q, the matrix P satisfies the *continuous Lyapunov equation* for A with forcing Q when

$$PA + A^{\mathsf{T}}P = -Q.$$

Reference: Hespanha, *Linear Systems Theory* (2nd ed.), Chapter 8, Theorem 8.2.

Definition 335 (Matrix quadratic form)**.** A real matrix P represents the quadratic form $x \mapsto x^{\mathsf{T}} P x$ on Euclidean state space.

Reference: standard quadratic Lyapunov-function construction.

Hespanha's Theorem 8.2 ties the spectral condition to solvability of $PA + A^{\mathsf{T}}P = -Q$. The direction a Lyapunov certificate is *used* in comes first: exhibiting a single positive-definite P proves the spectrum lies in the open left half-plane.

Theorem 336 (A Lyapunov certificate proves the spectrum lies in the open left half-plane)**.** *If P and Q are positive definite and $PM + M^{\mathsf{T}}P = -Q$, then every eigenvalue of M has strictly negative real part.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Complexify the equation and pair it against an eigenvector v of M with eigenvalue μ. The two terms contribute $\mu\, v^* P v$ and $\bar{\mu}\, v^* P v$, so the left-hand side is $2\,\mathrm{Re}(\mu)\, v^* P v$ while the right-hand side is $-v^* Q v$. Both $v^* P v$ and $v^* Q v$ are strictly positive (lemma 186), so $\mathrm{Re}(\mu) < 0$. □

Theorem 337 (The Lyapunov inequality certifies the Hurwitz condition)**.** *If P is positive definite and $-(PM + M^{\mathsf{T}}P)$ is positive definite, then M is Hurwitz (definition 315).*

This is Hespanha's Theorem 8.2, (5) ⇒ (3)*. It is stated through positive definiteness of the negated residual rather than through a matrix order relation, so that it needs no Loewner order.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Take the forcing to be minus the residual and apply theorem 336. □

The converse is a construction. One finite block of the integral $\int_0^m e^{tA^{\mathsf{T}}} Q e^{tA}\,\mathrm{d}t$ is summed into a convergent discrete Lyapunov series, with convergence and uniqueness both coming from the contraction $\|e^{Am}\| < 1$. A Hurwitz matrix supplies that contraction, which is theorem 339.

Theorem 338 (The Lyapunov equation from a contracting exponential)**.** *Suppose $\|e^{Am}\| < 1$ for some integer $m > 0$. Then for every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 334), and P is the unique matrix solution.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Write $L(P) = PA + A^{\mathsf{T}}P$ for the Lyapunov operator, $C = e^{Am}$, and $T(W) = C^{\mathsf{T}} W C$ for the congruence (Stein) operator.

One block. Let $R = \int_0^m e^{tA^{\mathsf{T}}} Q e^{tA}\,\mathrm{d}t$. Differentiating the integrand two ways and equating the derivatives by uniqueness gives the key identity $L(R) = T(Q) - Q$. R is positive definite because the integrand is, pointwise, and positive definiteness survives the integral.

Summing the blocks. $\|C\| < 1$ makes $\|T\| < 1$, so $\sum_k T^k R$ converges; call it P. It is positive definite, being a convergent sum of positive definite terms. Since A and C commute, L commutes with T, so $L(T^k R) = T^{k+1}Q - T^k Q$ and the series for $L(P)$ telescopes to $-Q$.

Uniqueness. A contraction has only the trivial fixed point, so L is injective; two solutions with the same forcing therefore coincide.

The contraction is taken in the Frobenius norm, the only place in the construction where $\|C^{\mathsf{T}}\| = \|C\|$ is needed. □

Theorem 339 (Lyapunov equation for a Hurwitz matrix)**.** *If a real matrix A is Hurwitz, then for every positive-definite Q there is a positive-definite matrix P satisfying $PA + A^{\mathsf{T}}P = -Q$, and this P is the unique matrix solution.*

omit [DecidableEq X] in Reference: Khalil, Nonlinear Systems*.*

Proof. A contracting integer-time matrix exponential is obtained from the Hurwitz spectrum. Integrating over one time block and summing the resulting discrete Lyapunov series constructs P; contraction proves convergence and uniqueness. □

Lemma 340 (From every forcing to one certificate)**.** *If every positive definite Q admits a positive definite solution of $PM + M^\mathsf{T}P = -Q$ (definition 334), then some positive definite P makes $-(PM + M^\mathsf{T}P)$ positive definite.*

This is Hespanha's Theorem 8.2, (4) ⇒ (5)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Instantiate the forcing at $Q = I$; the residual is then $-I$, whose negation is the identity. The edge carries no mathematics beyond the instantiation. □

8.2.5 The rate form

The Lyapunov inequality $PA + A^\mathsf{T}P \preceq -2cP$ carries an explicit decay rate: along a trajectory it gives $\dot{V} \leq -2cV$ for $V = x^\mathsf{T}Px$. It is not one of Hespanha's clauses, but it is the shape every contraction and guaranteed-decay-rate condition is written in, so it is recorded as a sixth equivalent.

Theorem 341 (The rate form certifies the Hurwitz condition)**.** *If P is positive definite, $c > 0$, and $PA + A^\mathsf{T}P \preceq -2cP$ — that is, $-(PA + A^\mathsf{T}P) - 2cP$ is positive semidefinite — then A is Hurwitz (definition 315).*

Along a trajectory this inequality gives $\dot{V} \leq -2cV$ for $V = x^\mathsf{T}Px$, so it certifies not merely stability but a decay rate.

Proof. Split $-(PA + A^\mathsf{T}P)$ as $(-(PA + A^\mathsf{T}P) - 2cP) + 2cP$: positive semidefinite plus positive definite is positive definite, so theorem 337 applies. □

Theorem 342 (A Hurwitz matrix admits a rate certificate)**.** *If A is Hurwitz (definition 315) then there are a positive definite P and a rate $c > 0$ with $PA + A^\mathsf{T}P \preceq -2cP$.*

Proof. Solve the Lyapunov equation for the *shifted* matrix. A Hurwitz A has a positive spectral margin c (theorem 318), so $A+cI$ is still Hurwitz (theorem 317). Solving $P(A+cI)+(A+cI)^\mathsf{T}P = -I$ (theorem 339) and expanding gives $PA+A^\mathsf{T}P = -I-2cP$, so the residual $-(PA+A^\mathsf{T}P)-2cP$ is *exactly* the identity.

No domination bound between two positive definite matrices is needed: the residual is not merely bounded below, it is I. □

8.2.6 Theorem 8.2

Theorem 343 (Hurwitz is solvability of the Lyapunov equation for every forcing)**.** *A real square matrix A is Hurwitz (definition 315) if and only if for every positive definite Q there is a positive definite P with $PA + A^\mathsf{T}P = -Q$ (definition 334), unique among all matrix solutions.*

This is Hespanha's Theorem 8.2, (3) ⇔ (4)*.*

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards is theorem 339. Backwards, instantiate the forcing at $Q = I$ and apply theorem 336. □

Theorem 344 (Hurwitz is the existence of one Lyapunov certificate)**.** *A real square matrix A is Hurwitz (definition 315) if and only if some positive definite P makes $-(PA + A^{\mathsf{T}}P)$ positive definite.*

This is Hespanha's Theorem 8.2, (3) $\Leftrightarrow$ (5), and the form in which a Lyapunov certificate is actually supplied: one matrix, not a family.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Forwards, theorem 339 produces a solution for each forcing and lemma 340 extracts one certificate from them; backwards is theorem 337. □

Theorem 345 (Hurwitz is the existence of a rate certificate)**.** *A real square matrix A is Hurwitz (definition 315) if and only if there are a positive definite P and a rate $c > 0$ with $PA+A^{\mathsf{T}}P \preceq -2cP$.*

This is clause 6 of theorem 346, which is not in Hespanha's list. It is the rate-carrying refinement of theorem 344, and the shape every contraction and guaranteed-decay-rate condition is written in.

Proof. theorem 342 and theorem 341. □

Theorem 346 (Hespanha's Theorem 8.2)**.** *For the time-invariant system $\dot{x} = Ax$ the following are equivalent.*

1. *The origin is asymptotically stable (definition 260).*
2. *The origin is exponentially stable (definition 265).*
3. *A is Hurwitz (definition 315).*
4. *For every positive definite Q there is a positive definite P with $PA + A^{\mathsf{T}}P = -Q$ (definition 334), and this P is the unique matrix solution.*
5. *There is a positive definite P with $-(PA + A^{\mathsf{T}}P)$ positive definite.*
6. *There are a positive definite P and a rate $c > 0$ with $PA + A^{\mathsf{T}}P \preceq -2cP$.*

Clause 6 is not in Hespanha's list. It is the rate-carrying form of clause 5: along a trajectory it gives $\dot{V} \leq -2cV$ for $V = x^{\mathsf{T}}Px$, hence $V(t) \leq V(0)e^{-2ct}$, and it is the shape every contraction and guaranteed-decay-rate condition is written in.

Reference: Hespanha, Linear Systems Theory *(2nd ed.), Chapter 8, Theorem 8.2.*

Proof. Every clause is tied to clause 3 and the cycle is closed by transitivity: $1 \Leftrightarrow 3$ is theorem 332, $2 \Leftrightarrow 3$ is theorem 333, $3 \Leftrightarrow 4$ is theorem 343, $3 \Leftrightarrow 5$ is theorem 344, and $3 \Leftrightarrow 6$ is theorem 345. □

8.2.7 Quadratic forms along a linear vector field

What makes a solution of the Lyapunov equation a Lyapunov *function*: the forcing matrix is exactly the dissipation rate of the associated quadratic form.

Theorem 347 (A Lyapunov solution makes its quadratic form decrease)**.** *Let P solve $PA+A^{\mathsf{T}}P = -Q$ (definition 334). Then the derivative of the centered quadratic form $x \mapsto (x - x_{\text{eq}})^{\mathsf{T}}P(x - x_{\text{eq}})$ (definition 335) along the linear vector field $y \mapsto Ay$ is*

$$-(x - x_{\text{eq}})^{\mathsf{T}}Q\,(x - x_{\text{eq}}).$$

This is what makes a Lyapunov solution a Lyapunov function*: the forcing Q is exactly the dissipation rate.*

Proof. The derivative of the quadratic form in the direction Ay is $\langle y, Py'\rangle + \langle y', Py\rangle$ with $y' = Ay$, which is the form represented by $PA + A^{\mathsf{T}}P$. Transporting the equation through `toEuclideanCLM` — under which transpose becomes adjoint — rewrites that as $-Q$. □

Lemma 348 (Identity forcing gives decrease at the squared norm)**.** *For P solving $PA+A^{\mathsf{T}}P = -I$ (definition 334), the derivative of the centered quadratic form (definition 335) along $y \mapsto Ay$ is $-\|x - x_{\mathrm{eq}}\|^2$.*

Proof. theorem 347 at $Q = I$, where the quadratic form is the squared Euclidean norm. □

8.2.8 A quadratic certificate for an unstable eigenmode

The instability counterpart (theorem 355): an eigenvalue with positive real part yields a real quadratic Chetaev certificate. The construction solves a shifted Lyapunov–Sylvester equation, whose injectivity rests on a nonresonance condition on the spectrum — which is what the lemmas below supply.

Lemma 349 (A nonresonant bilinear form vanishes on a pair of generalized eigenspaces)**.** *Let T be an endomorphism of a complex vector space and B a bilinear form satisfying the Sylvester-type identity*

$$B(x, Ty) + B(Tx, y) = c\, B(x, y) \qquad \forall\, x, y.$$

If $\xi + \nu \neq c$, then B vanishes on $\ker(T-\xi)^k \times \ker(T-\nu)^l$ for all k, l.

Proof. Strong induction on $k + l$. The base cases are the zero generalized eigenspaces, where the vector is 0. Otherwise write $Tx = N_x + \xi x$ and $Ty = N_y + \nu y$ with N_x, N_y one order lower; the induction hypothesis kills $B(N_x, y)$ and $B(x, N_y)$, leaving $(\xi + \nu - c)B(x, y) = 0$, and the coefficient is nonzero by nonresonance. □

Lemma 350 (A nonzero generalized eigenvector witnesses its eigenvalue)**.** *If $x \neq 0$ lies in the maximal generalized eigenspace of T at ξ, then ξ is an eigenvalue of T.*

Proof. Membership gives $x \in \ker(T - \xi)^k$ for some k, necessarily $k \neq 0$ since $x \neq 0$; so that generalized eigenspace is nontrivial, and a nontrivial generalized eigenspace forces an eigenvalue. □

Lemma 351 (A nonresonant bilinear form vanishes identically)**.** *Let V be a finite-dimensional complex vector space, T an endomorphism, and B a bilinear form with $B(x, Ty) + B(Tx, y) = c\, B(x, y)$. If no pair of eigenvalues of T sums to c, then $B = 0$.*

This is the uniqueness half of solvability for the Sylvester-type equation: the associated operator has trivial kernel exactly when the spectrum is nonresonant.

Proof. Over $\mathbb{C}$ the generalized eigenspaces of T span V, so it suffices to check $B(x, y) = 0$ for x, y in generalized eigenspaces. A nonzero such vector witnesses its eigenvalue (lemma 350), so nonresonance applies and lemma 349 gives $B(x, y) = 0$. □

Lemma 352 (Right multiplication acts on the right argument)**.** *For square matrices H, A over a commutative semiring, $(HA)(x, y) = H(x, Ay)$ as bilinear forms.*

Proof. Both sides unfold to $x \cdot (HAy)$ by associativity of matrix-vector multiplication. □

Lemma 353 (Left multiplication by the transpose acts on the left argument)**.** *For square matrices H, A over a commutative semiring, $(A^{\mathsf{T}}H)(x, y) = H(Ax, y)$ as bilinear forms.*

Together with lemma 352 this is what turns the matrix equation $HA + A^{\mathsf{T}}H = cH$ into the Sylvester-type identity on bilinear forms that lemma 351 consumes.

Proof. Move the transpose across the dot product: $x \cdot (A^{\mathsf{T}}Hy) = (Ax) \cdot (Hy)$. □

Lemma 354 (A symmetric matrix represents a symmetric form)**.** *If H is a symmetric real matrix then $H(x, y) = H(y, x)$ as a bilinear form.*

Proof. Move the transpose across the dot product and apply $H^{\mathsf{T}} = H$. □

Theorem 355 (Quadratic instability certificate from an unstable eigenpair)**.** *A complex eigenpair of a real matrix A with $\operatorname{Re}\mu > 0$ yields a Hermitian real matrix H, a shift $\alpha > 0$, and a direction w such that $w^{\mathsf{T}}Hw > 0$ and*

$$HA + A^{\mathsf{T}}H - 2\alpha H$$

is positive definite.

Reference: Hahn, Stability of Motion*; Khalil,* Nonlinear Systems*.*

Proof. Choose a shift avoiding the finitely many pairwise spectral resonances, solve the resulting Lyapunov–Sylvester equation by finite-dimensional injectivity, and extract w from the real or imaginary part of the supplied eigenvector. □

Chapter 9

Lyapunov's indirect method

Near an equilibrium a nonlinear system behaves like its linearization: if the Jacobian is Hurwitz the equilibrium is locally exponentially stable, and if the Jacobian has an eigenvalue with positive real part it is unstable. Both branches run the same way. A quadratic certificate for the linear system (theorem 339, theorem 355) is a certificate for the nonlinear one as well, once the linearization error is absorbed near the equilibrium.

9.1 The linearization error

Theorem 356 (Quantitative linearization error bound)**.** *Let A be the Fréchet derivative of f at x_0. For every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\|x - x_0\| < \delta \implies \|f(x) - f(x_0) - A(x - x_0)\| \le \varepsilon \|x - x_0\|.$$

This is the ε–δ form of $f(x) - f(x_0) - A(x - x_0) = o(\|x - x_0\|)$.

Proof. □

Lemma 357 (Absorbing the linearization remainder)**.** *Let f be C^1 with $Df(x_{\mathrm{eq}}) = A$, let M be a matrix, and let $c > 0$. Then there is $r > 0$ such that for every x with $\|x - x_{\mathrm{eq}}\| < r$,*

$$\left| D\, q_M(x)\, [\, f(x) - f(x_{\mathrm{eq}}) - A(x - x_{\mathrm{eq}})\,] \right| \le c\, \|x - x_{\mathrm{eq}}\|^2,$$

where q_M is the centred quadratic form $q_M(x) = \langle M(x - x_{\mathrm{eq}}),\, x - x_{\mathrm{eq}} \rangle$.

The bracket is the first-order remainder of f at x_{eq}. The point is that c is arbitrary*: near enough to the equilibrium the remainder perturbs $\dot q_M$ by less than any prescribed multiple of $\|x - x_{\mathrm{eq}}\|^2$, which is the order of $\dot q_M$ itself. This is why the linearization decides stability.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7 (the quadratic-Lyapunov proof of Lyapunov's indirect method).*

Proof. Write $y = x - x_{\mathrm{eq}}$ and e for the remainder. Differentiability of f at x_{eq} gives, for any $\eta > 0$, a radius within which $\|e\| \le \eta \|y\|$. The derivative of a centred quadratic form is bounded by $|Dq_M(x)[e]| \le 2\|M\|\, \|y\|\, \|e\|$, so the product is at most $2\|M\| \eta\, \|y\|^2$. Choosing $\eta = c/(4(\|M\| + 1))$ makes $2\|M\|\eta \le c$. □

9.2 The stable branch

The stable-branch theorem gives uniform local exponential decay on every finite forward solution segment, without silently excluding solutions that are not globally defined.

Theorem 358 (Lyapunov's indirect method, stable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1, and let x_{eq} be an equilibrium. If its Jacobian A at the equilibrium is Hurwitz, then x_{eq} is locally exponentially stable (definition 265) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems*.*

Proof. Solve the identity-forced Lyapunov equation for a positive-definite P; after absorbing the $o(\|x - x_{\mathrm{eq}}\|)$ linearization remainder on a small ball, $V(x) = (x - x_{\mathrm{eq}})^{\mathsf{T}} P (x - x_{\mathrm{eq}})$ is an exponential Lyapunov function with exponent 2 there, so theorem 299 applies. □

9.3 The unstable branch

The quadratic certificate is fed to Chetaev's theorem (theorem 306).

Theorem 359 (Lyapunov's indirect method, unstable branch)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 with $f(x_{\mathrm{eq}}) = 0$. If the Jacobian at x_{eq} has a complex eigenvalue with positive real part, then x_{eq} is unstable (definition 257) for any system whose state equation is $\dot{x} = f(x)$.*

Reference: Khalil, Nonlinear Systems *(Lyapunov's indirect method).*

Proof. Build the real quadratic certificate supplied by theorem 355, absorb the first-order nonlinear remainder on a small ball, and apply the exponential Chetaev criterion. □

Theorem 360 (Instability of a linear system)**.** *If $A \in \mathbb{R}^{n \times n}$ has a complex eigenvalue μ with $\operatorname{Re} \mu > 0$, then the origin is unstable (definition 257) for any system on $\mathbb{R}^n$ whose state equation is $\dot{x} = Ax$.*

Reference: Khalil, Nonlinear Systems *(3rd ed.), Theorem 4.7.*

Proof. The field $x \mapsto Ax$ is C^1, vanishes at 0 and is its own Jacobian, so this is theorem 359 at $x_{\mathrm{eq}} = 0$. □

Chapter 10

Controllability and observability

10.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \qquad y = Cx,$$

with state matrix A, input matrix B, and output matrix C, this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

10.1.1 Observability matrix

Definition 361 (Observability matrix)**.** The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n\times n}$ and $C \in \mathbb{F}^{p\times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C\,A \\ C\,A^2 \\ \vdots \\ C\,A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np)\times n}.$$

Rows are indexed by $\mathrm{Fin}\, n \times \mathrm{Fin}\, p$, so that A^k is available without casting $k : \mathrm{Fin}\, n$ through Fin.val.

Definition 362 (Observability)**.** A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C\,A^k\,x = 0 \qquad \text{for every } k = 0, 1, \ldots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge theorem 368 has real content.

Lemma 363 (Entries of the observability matrix)**.** *Block-row entry shape: at row (k, i) and column j, the observability matrix coincides with the (i, j) entry of $C\,A^k$:*

$$\mathcal{O}(A, C)_{(k,i),\,j} = (C\,A^k)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 361. □

Lemma 364 (Action of the observability matrix)**.** *For every state $x \in \mathbb{F}^n$ and every $(k, i) \in \operatorname{Fin} n \times \operatorname{Fin} p$,*

$$(\mathcal{O}(A, C) \cdot x)_{(k,i)} = (C\, A^k \cdot x)_i.$$

This is the definitional bridge that powers theorem 368.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see definition 361). □

10.1.2 Controllability matrix

Definition 365 (Controllability matrix)**.** The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = \begin{bmatrix} B & A\,B & A^2\,B & \cdots & A^{n-1}\,B \end{bmatrix} \in \mathbb{F}^{n\times(nm)}.$$

Columns are indexed by $\operatorname{Fin} n \times \operatorname{Fin} m$, so that A^k is available without casting $k : \operatorname{Fin} n$ through Fin.val.

Lemma 366 (Entries of the controllability matrix)**.** *Block-column entry shape: at row i and column (k, j), the controllability matrix coincides with the (i, j) entry of $A^k\,B$:*

$$\mathcal{C}(A, B)_{i,\,(k,j)} = (A^k\,B)_{i,\,j}.$$

Proof. Holds definitionally by the encoding chosen for definition 365. □

Definition 367 (Controllability)**.** A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \ldots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k\,B\,u_k.$$

This phrasing does not name the controllability matrix, so the bridge theorem 370 has real content.

10.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 368 (Observability via a trivial kernel)**.** *A finite-dimensional linear system (A, C) is observable in the sense of definition 362 if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication:*

$$\operatorname{IsObservable}(A, C) \iff (\forall x,\ \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0).$$

Proof. Both directions follow from the bridge lemma 364, which identifies the (k, i)-coordinate of $\mathcal{O}(A, C) \cdot x$ with the i-coordinate of $(C\,A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. □

10.1.4 Rank characterizations

Theorem 369 (Observability rank test)**.** *A finite-dimensional system (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e.*

$$\operatorname{rank} \mathcal{O}(A, C) = n,$$

where n is the state dimension.

Proof. Chain the kernel-form milestone theorem 368 with the matrix-level bridge $(\forall x,\ M \cdot x = 0 \Rightarrow x = 0) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{O}(A, C)$. □

Theorem 370 (Controllability rank test)**.** *A linear system (A, B) is controllable if and only if the controllability matrix has full row rank,*

$$\operatorname{rank} \mathcal{C}(A, B) = n,$$

where n is the state dimension.

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y,\ \exists x,\ M\,x = y) \iff \operatorname{rank} M = n$ from 'MatrixAlgebra.Rank', applied with $M = \mathcal{C}(A, B)$. □

10.1.5 Hautus observability

Definition 371 (Unobservable subspace)**.** The *unobservable subspace* of (A, C) is the A-invariant subspace

$$\mathcal{N}(A, C) \;=\; \{\, v \in \mathbb{C}^n \;:\; C\,A^k\,v = 0 \text{ for every } k = 0, 1, \dots, n-1 \,\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge theorem 372 reads as content.

Theorem 372 (Observability via the unobservable subspace)**.** *A finite-dimensional system (A, C) is observable in the sense of definition 362 if and only if its unobservable subspace is trivial:*

$$\mathcal{N}(A, C) = \{0\} \iff \text{IsObservable}(A, C).$$

Proof. Both directions are membership unfoldings of definition 371 against definition 362. □

Lemma 373 (A-invariance of the unobservable subspace)**.** *The unobservable subspace is closed under the action of A: for every $v \in \mathcal{N}(A, C)$, also $A\,v \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $C\,A^n\,v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$-linear combination of $C\,A^i\,v$ for $i < n$. Each of those is zero, so the combination is zero. □

Definition 374 (Hautus observability matrix)**.** The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) \;=\; \begin{bmatrix} \mu I - A \\ C \end{bmatrix} \in \mathbb{C}^{(n+p)\times n}.$$

Lemma 375 (Kernel of the Hautus matrix)**.** *For every $\mu \in \mathbb{C}$ and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (A\,v = \mu\,v \ \wedge\ C\,v = 0).$$

Proof. □

Theorem 376 (Unobservability gives a Hautus rank drop)**.** *If (A, C) is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$*.*

Proof. The unobservable subspace definition 371 is nontrivial by theorem 372. It is A-invariant by lemma 373, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via lemma 375. □

Theorem 377 (A Hautus rank drop gives unobservability)**.** *If $\mu \in \mathbb{C}$ and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k, $A^k v = \mu^k v$, so $C\,A^k\,v = \mu^k\,Cv = 0$ for every k. The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. □

Theorem 378 (Hautus (PBH) test for observability)**.** *A finite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\mathrm{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C},\ \ker H_{A,C}(\mu) = \{0\}.$$

Proof. Combine theorem 376 (failure direction) and theorem 377 (converse). □

10.1.6 Hautus controllability

Definition 379 (Hautus controllability matrix)**.** The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H^{\mathrm{ctrl}}_{A,B}(\mu) \;=\; [\mu I - A \quad B] \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 380 (Hautus (PBH) test for controllability)**.** *A finite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H^{\mathrm{ctrl}}_{A,B}(\mu)$ has full row rank:*

$$\mathrm{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C},\ \mathrm{rank}\, H^{\mathrm{ctrl}}_{A,B}(\mu) = n.$$

Proof. Combine the duality bridge `isControllable_iff_isObservable_transpose` with the observability Hautus iff theorem 378, then convert the kernel-form RHS to the rank form via the matrix bridge `mulVec_kernel_trivial_iff_rank_eq_card_cols` from 'MatrixAlgebra.Rank', and finally identify the transposed Hautus matrices. □

10.2 Reachable subspace and Kalman decomposition

Definition 381 (Reachable subspace)**.** The reachable subspace of a pair (A, B) is the column span of its controllability matrix:

$$\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B).$$

Thus it is exactly the span of $B, AB, \dots, A^{n-1}B$.

Reference: Hespanha, *Linear Systems Theory.*

Lemma 382 (Membership in the reachable subspace)**.** *Let $A \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times m}$, and let $\mathcal{C}(A, B) \in \mathbb{F}^{n\times(nm)}$ be their controllability matrix. For $x \in \mathbb{F}^n$,*

$$x \in \mathcal{R}(A, B) \quad \iff \quad \exists u : \operatorname{Fin}(n) \times \operatorname{Fin}(m) \to \mathbb{F}, \qquad \mathcal{C}(A, B)u = x.$$

Here $\mathcal{R}(A, B) = \operatorname{range} \mathcal{C}(A, B)$ is the reachable subspace.

Proof. □

Theorem 383 (Controllability via the reachable subspace)**.** *A pair (A, B) is controllable if and only if its reachable subspace is the whole state space.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Lemma 384 (A-invariance of the reachable subspace)**.** *The reachable subspace is A-invariant: $A\mathcal{R}(A, B) \subseteq \mathcal{R}(A, B)$.*

Reference: Hespanha, Linear Systems Theory*.*

Proof. □

Definition 385 (Kalman decomposition)**.** A Kalman decomposition of (A, B, C) records four coordinate sectors adapted to the reachable and unobservable subspaces. Their direct sums recover those subspaces and the full state space. The chosen complements are noncanonical and are not individually asserted to be invariant under A.

References: Hespanha, *Linear Systems Theory*; Kailath, *Linear Systems.*

Theorem 386 (Existence of the Kalman decomposition)**.** *Every finite-dimensional complex state-space system has four coordinate sectors adapted to its reachable and unobservable subspaces. The complements are noncanonical vector-space complements and are not individually asserted to be invariant under A.*

References: Hespanha, Linear Systems Theory*; Kailath,* Linear Systems*.*

Proof. □

Definition 387 (Adapted coordinate equivalence)**.** Addition of the four Kalman coordinate sectors defines a linear equivalence $X_{c\bar{o}} \times X_{co} \times X_{\bar{c}\bar{o}} \times X_{\bar{c}o} \simeq \mathbb{C}^n$. Choosing sector bases therefore gives an adapted basis of the original state space.

Theorem 388 (Block-zero pattern in adapted coordinates)**.** *In a basis adapted to the four Kalman coordinate sectors, ordered as controllable-unobservable, controllable-observable, uncontrollable-unobservable, uncontrollable-observable, the system has the forced-zero pattern*

$$A' = \begin{bmatrix} * & * & * & * \\ 0 & * & 0 & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix}, \qquad B' = \begin{bmatrix} * \\ \\ 0 \\ 0 \end{bmatrix}, \qquad C' = \begin{bmatrix} 0 & * & 0 & * \end{bmatrix}.$$

Every displayed zero is asserted entrywise; no vanishing claim is made for the starred blocks. The chosen complement sectors are not individually asserted to be invariant under A.

References: Hespanha, Linear Systems Theory*; Kalman, "Mathematical Description of Linear Dynamical Systems" (1963); Kailath,* Linear Systems*.*

Proof. Reachable-subspace invariance forces the lower-left state and input zeros, unobservable-subspace invariance forces the remaining state zeros, and the zeroth observability condition forces the output zeros. Coordinate-basis representation lemmas translate these component statements to individual matrix entries. □